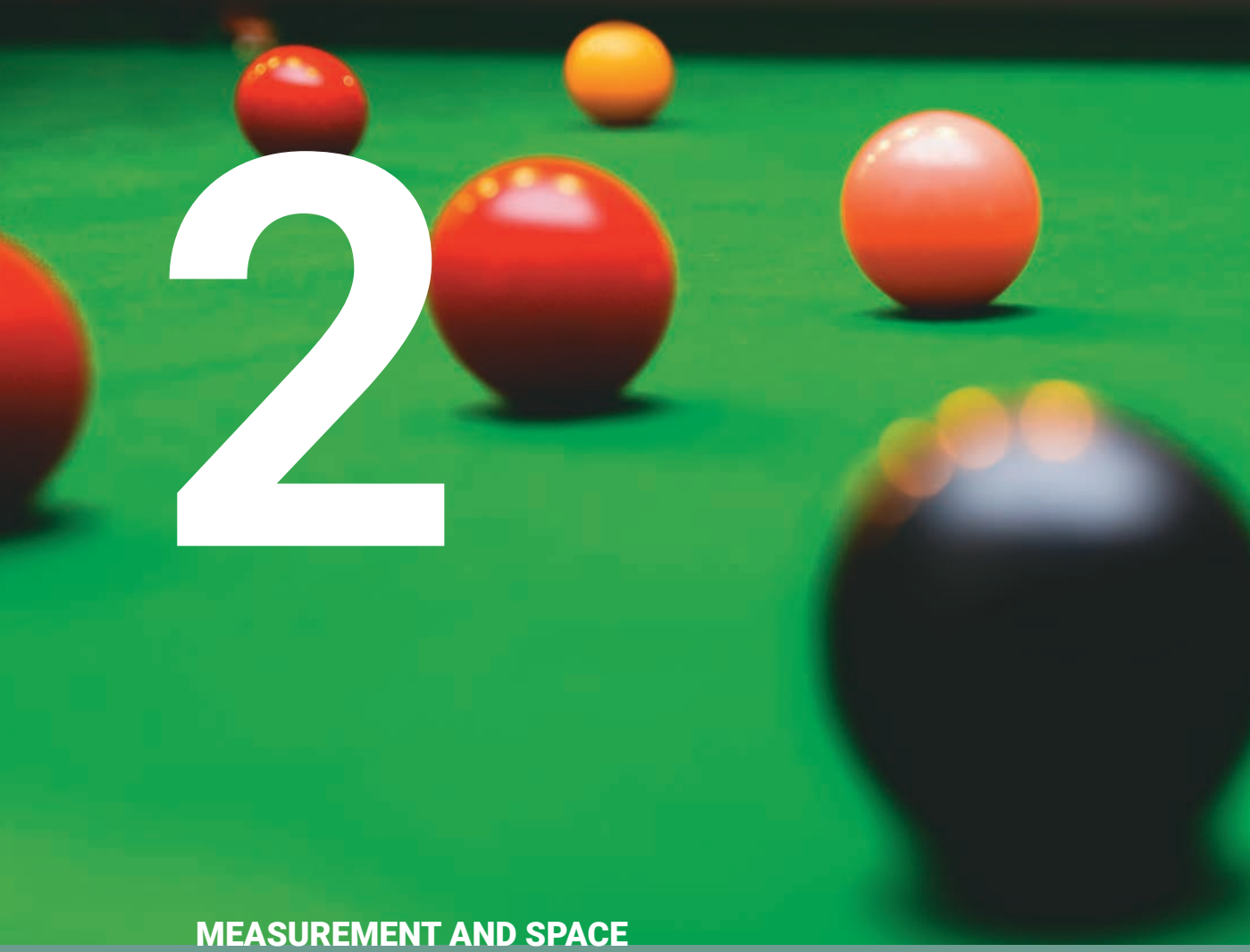


2

MEASUREMENT AND SPACE

ANGLES

Angles are used in all aspects of life. Builders and engineers use angles for buildings, roads and bridges. Sportspeople use angles when playing snooker, when shooting for goals and even when skiing. Artists use their knowledge of angles to draw portraits and paintings.



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Chapter outline

	Working mathematically				
2.01 Naming angles	U	F		R	C
2.02 Measuring and drawing angles	U	F		R	C
2.03 Types of angles	U	F		R	C
2.04 Complementary and supplementary angles	U	F		R	C
2.05 Angles at a point and vertically opposite angles	U	F		R	C
2.06 Parallel and perpendicular lines	U	F		R	C
2.07 Corresponding angles on parallel lines	U	F		R	C
2.08 Alternate angles on parallel lines	U	F		R	C
2.09 Co-interior angles on parallel lines	U	F		R	C
2.10 Angles on parallel lines	U	F		R	C
2.11 Proving parallel lines	U	F		R	C

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Wordbank

alternate angles Pairs of angles on opposite sides of the transversal crossing 2 lines, between the lines

co-interior angles Pairs of angles on the same side of the transversal crossing 2 lines, between the lines

complementary angles 2 angles that add to 90°

corresponding angles Pairs of 'matching' angles formed when a transversal crosses 2 lines

obtuse angle An angle whose size is between 90° and 180°

parallel lines Lines that point in the same direction and do not intersect

perpendicular lines Lines that intersect at right angles

supplementary angles 2 angles that add to 180°

transversal A line that cuts across 2 or more other lines

Quiz
Wordbank 2

Videos (6):

2.02 Naming angles
2.04 Measuring angles
2.05 Angles at a point and vertically opposite angles • Angle geometry
2.07, 2.08, 2.09 Angle relationships
2.10 Angles on parallel lines

Quizzes (5):

- Wordbank 2
- SkillCheck 2
- Mental skills 2
- Language of maths 2
- Test yourself 2

Skillsheets (3):

2.02 Measuring angles
2.03 Types of angles
2.07, 2.08, 2.09 Angles and parallel lines

Worksheets (11):

2.01 A page of angles
2.02 A page of protractors • Using a protractor • Estimating and measuring angles • A page of angles
2.03 A page of angles
2.05 Straight angles, right angles and revolutions
2.06 A page of intervals
2.07, 2.08, 2.09 Investigating angles on parallel lines
2.09 Angles in parallel lines
2.10 Find the unknown angle • What is the diagram? • Matching angles
Mind map Angles

Puzzles (5):

2.01, 2.02, 2.03 Angle cards
2.05 Angles: a dog day • Angles
2.09 Angles in parallel lines
Language of maths Angles crossword

Presentation (1):

2.02 Measuring angles with a protractor

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

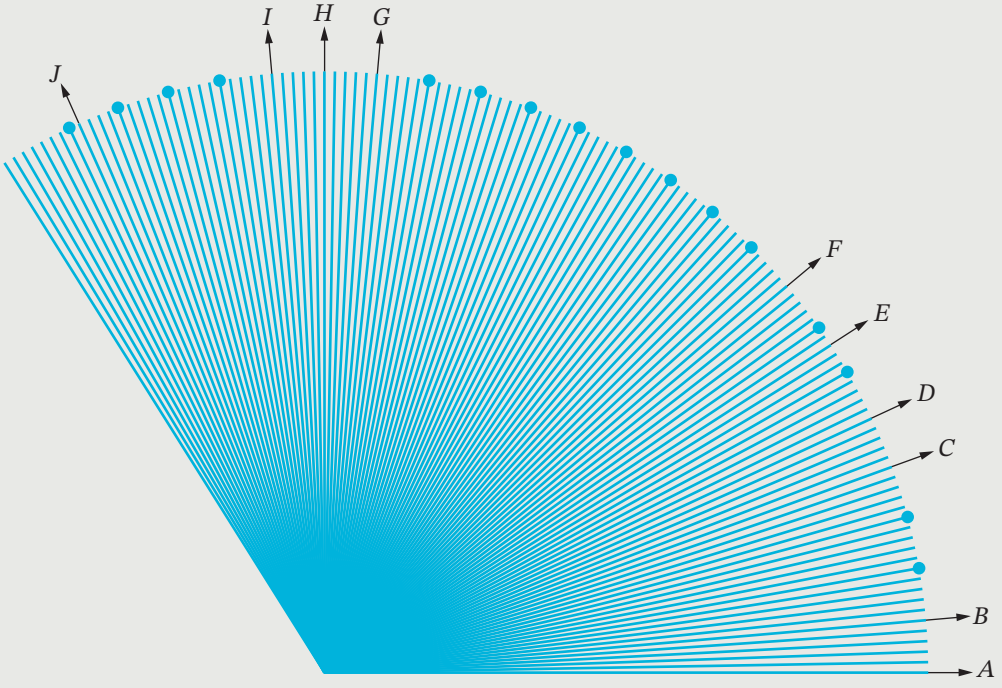
In this chapter you will:

- learn about points, lines, rays and intervals
- use 1 or 3 letters to name and label an angle
- estimate and measure the size of an angle
- classify angles such as acute, right, obtuse, straight, reflex and revolution
- name pairs of angles that are adjacent, complementary, supplementary and vertically opposite
- solve problems involving angles on a straight line, angles at a point and vertically opposite angles
- identify corresponding, alternate and co-interior angles when parallel lines are crossed by a transversal, and their properties
- use the above properties to solve geometry problems and to prove that 2 lines are parallel
- identify and draw parallel and perpendicular lines

SkillCheck ANSWERS ON P. 563

Quiz
SkillCheck 2

All questions refer to this angle diagram, where each gap represents 1° of angle size.



1 What is the angle size between the lines I and J? Select the correct answer **A**, **B**, **C** or **D**.

A 19° **B** 29° **C** 34° **D** 39°

2 What is the angle size between the lines labelled:

a A and C? **b** A and D? **c** B and C?

d C and F? **e** A and F? **f** B and G?

g D and G? **h** E and H? **i** D and I?

j C and J? **k** B and E? **l** E and J?

3 Find one pair of labelled lines that have a 45° angle between them.

4 Find 2 pairs of labelled lines that have a 90° angle between them.

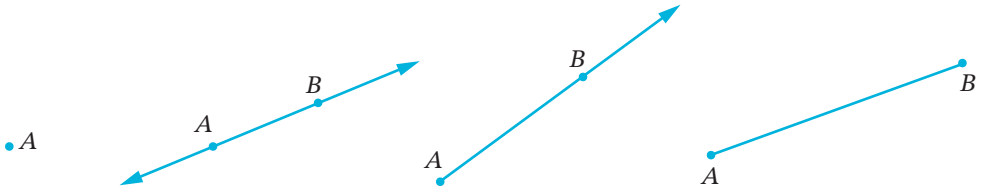
5 Find the pair of labelled lines that have each angle size between them.

a 7° **b** 8° **c** 13° **d** 28°

e 50° **f** 89° **g** 95° **h** 114°

2.01 Naming angles

Points and lines



The point A | The line AB | The ray AB | The interval AB

In geometry, a **point** is a position in space, represented by a dot and named by a capital letter such as A.

A **line** is straight and infinite in both directions, represented by arrows in both directions. It has no length or thickness, and is named by 2 points on the line, such as AB.

A **ray** is 'half a line', infinite in one direction only. It has a point at one end and an arrow at the other end.

An **interval** is 'part of a line' that has a point at each end (no arrows). It is also called a **line segment** and does have a length (but no thickness).

Angles

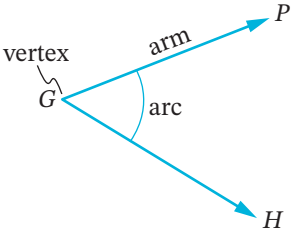
Worksheet
A page of angles

Puzzle
Angle cards

An **angle** describes the size of a turn or rotation. It is drawn with 2 rays called **arms** that meet at a **vertex**. The size of the turn is normally marked with a curve called an **arc**. The angle marked in this diagram can be written as:

$\angle G$, $\angle PGH$ or $\angle HGP$

The middle letter always labels the vertex of the angle.

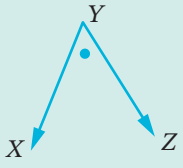


Video
Naming angles

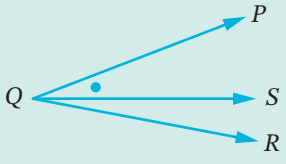
Example 1

Name the angle marked with • in each diagram.

a



b



Solution

a $\angle Y$ or $\angle XYZ$ or $\angle ZYX$

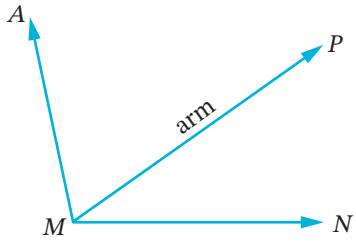
b $\angle PQS$ or $\angle SQP$

Note: We cannot use $\angle Q$ in part **b** because it is not clear which angle that means. There are 3 different angles whose vertex is $\angle Q$. They are $\angle PQS$, $\angle SQR$ and $\angle PQR$.

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Adjacent angles

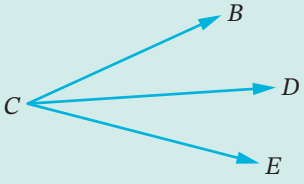
In this diagram, $\angle AMP$ and $\angle PMN$ share a common arm, PM. Angles that are next to each other in this way are called **adjacent angles**.



2.01

Example 2

What angle is adjacent to $\angle DCE$?



Video
Naming angles

Solution

$\angle BCD$ and $\angle DCE$ share a common arm CD. So $\angle BCD$ is adjacent to $\angle DCE$.

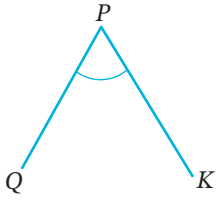
EXERCISE 2.01 ANSWERS ON P. 563

Naming angles UFRC

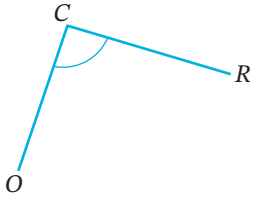
EXAMPLE 1

1 Name each angle in 2 different ways. C

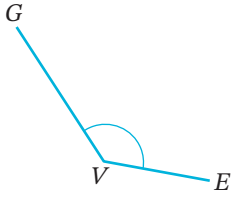
a



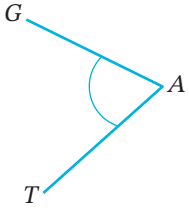
b



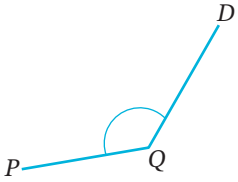
c



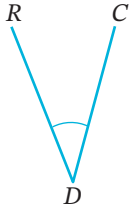
d



e



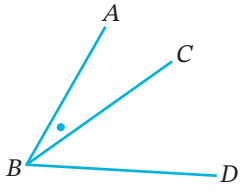
f



2 How can the angle marked • be named? Select the correct answer **A**, **B**, **C** or **D**. C

A $\angle ABD$ **B** $\angle CBD$

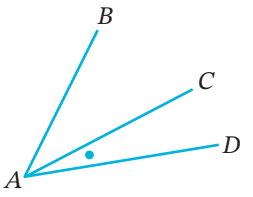
C $\angle ABC$ **D** $\angle BCA$



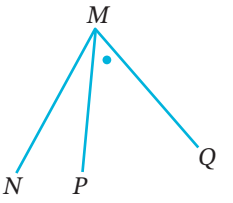
Foundation Standard Complex

3 Name the angle marked • in each diagram. C

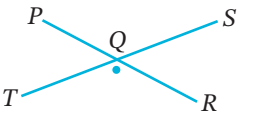
a



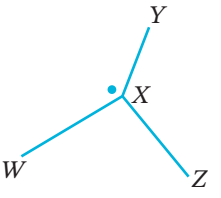
b



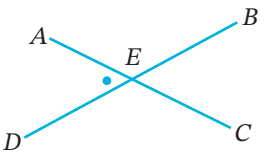
c



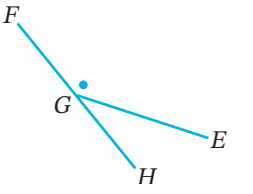
d



e



f



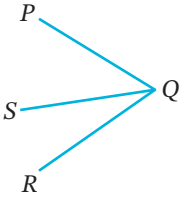
4 Draw each angle named, labelling it correctly.

a $\angle POT$ **b** $\angle TAF$ **c** $\angle AFE$ **d** $\angle H$

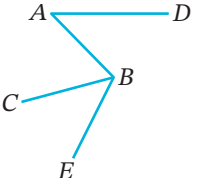
EXAMPLE 2

5 Name the angle adjacent to: C

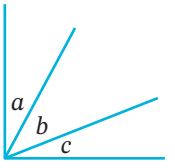
a $\angle RQS$



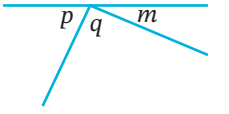
b $\angle ABC$



c the angle marked c



d the angle marked p

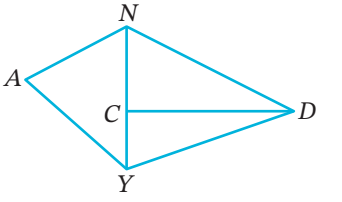


6 **a** There are 13 different angles on this diagram. Name them all. R C

b Name all 3 angles that have C as their vertex.

c What type of angle is $\angle NCY$?

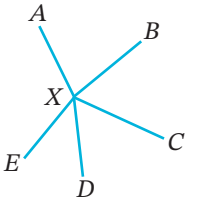
d Name the angle that is adjacent to $\angle YND$.



7 Which angle is adjacent to $\angle AXB$?
Select the correct answer **A**, **B**, **C** or **D**.

A $\angle BXC$ **B** $\angle DXE$

C $\angle DXC$ **D** $\angle CXE$

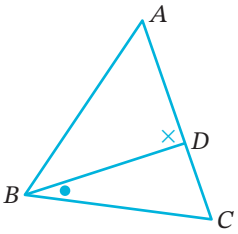


Foundation Standard Complex

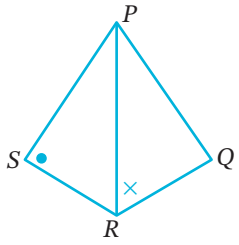
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8 Name the angles marked • and × in each diagram. C

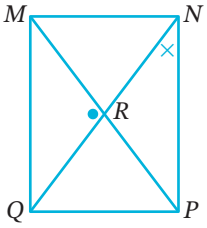
a



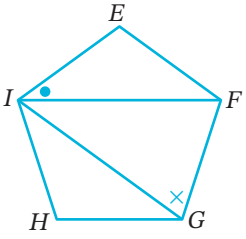
b



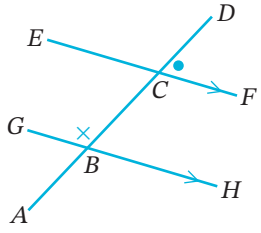
c



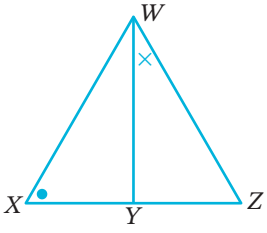
d



e



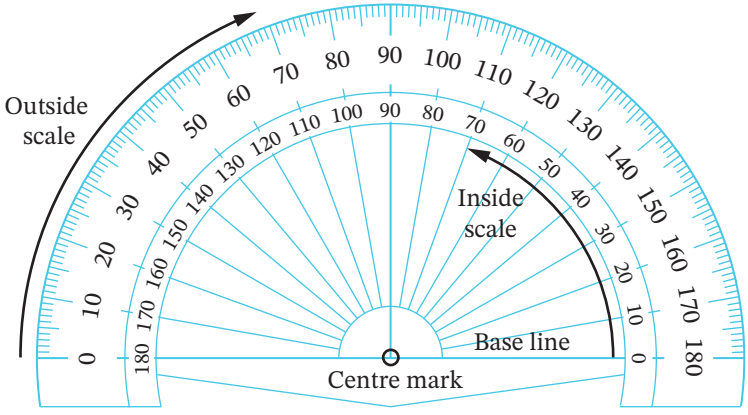
f



9 For each diagram in question **8**, name a pair of adjacent angles. R C

Measuring and drawing angles 2.02

A **protractor** is an instrument used for measuring and drawing angles.



Skillsheet
Measuring angles

Presentation
Measuring angles with a protractor

Worksheets
A page of protractors

Using a protractor

Estimating and measuring angles

Foundation Standard Complex

Video
Measuring angles

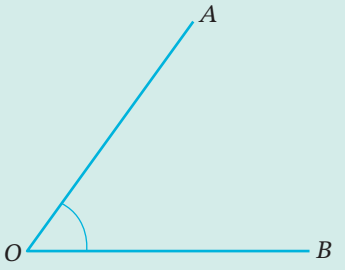
Worksheet
A page of angles

Puzzle
Angle cards

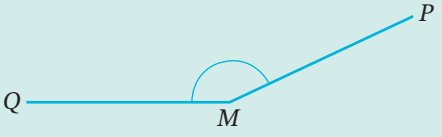
Example 3

Use a protractor to measure the size of each marked angle.

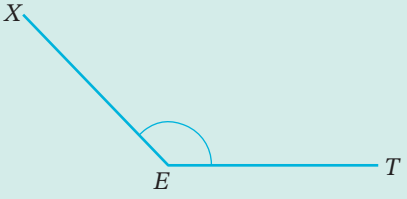
a



b



c

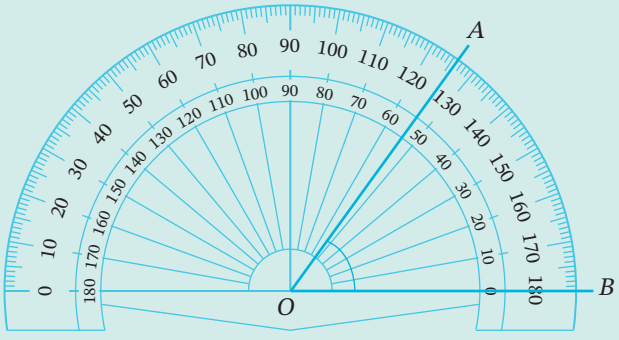


Solution

a Measuring $\angle AOB$:

- Line up OB with the base line of the protractor.
- Place the centre mark over the vertex, O.
- The angle is smaller than 90°.
- Use the inside scale, counting from 0°.

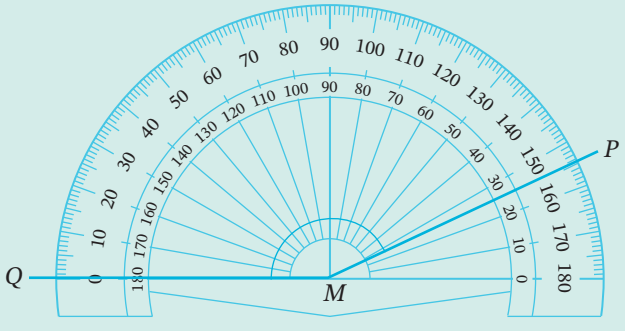
$\angle AOB = 54°$



b Measuring $\angle PMQ$:

- Line up QM with the base line of the protractor.
- Place the centre mark over the vertex, M.
- The angle is greater than 90°.
- Use the outside scale, counting from 0°.

$\angle PMQ = 155°$

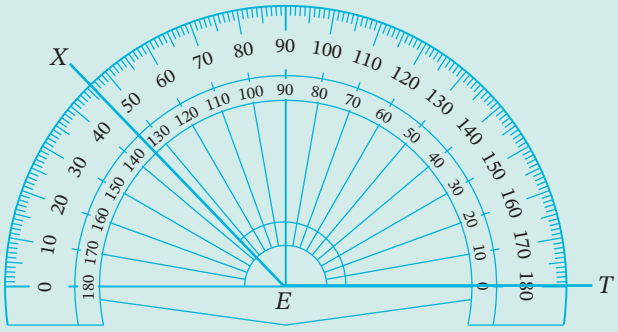


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c Measuring $\angle TEX$:

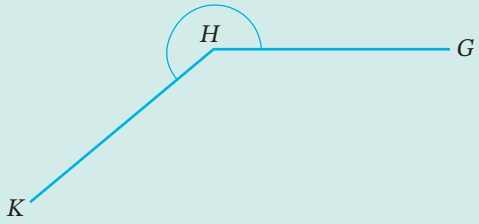
- Line up TE with the base line of the protractor.
- Place the centre mark over the vertex, E.
- $\angle TEX$ is bigger than 90°.
- Use the inside scale.

$\angle TEX = 134°$



Example 4

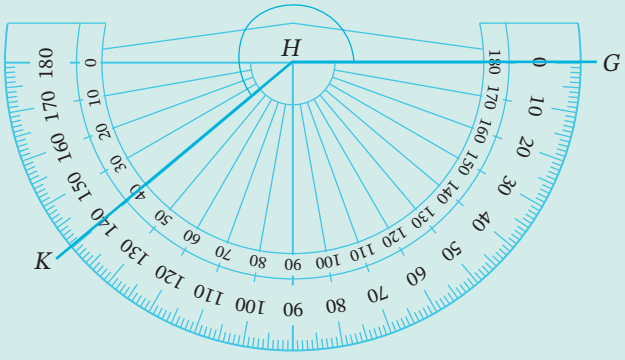
Measure the **reflex angle** GHK.



Solution

As the protractor only measures up to 180°, turn it upside down to measure the smaller adjacent angle underneath first.

Obtuse $\angle GHK = 140°$



To calculate the size of reflex $\angle GHK$, subtract 140° from 360° as there are 360° in a revolution.

Reflex $\angle GHK = 360° - 140° = 220°$

2.02

Video
Measuring angles

Example 5

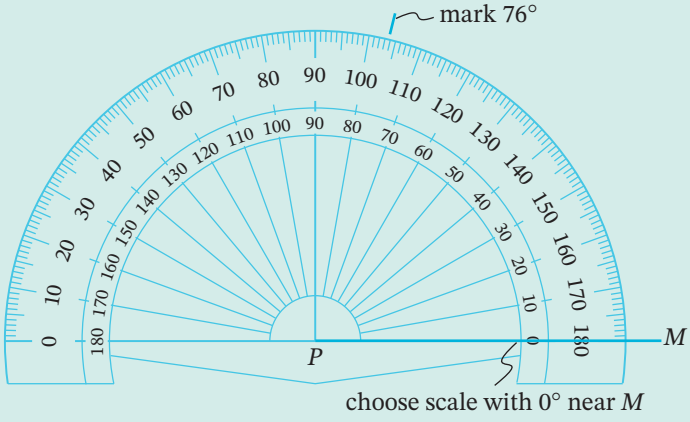
Construct $\angle KPM$ of size 76°.

Solution

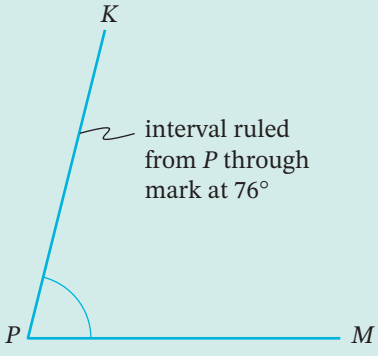
- Draw an interval with endpoints P and M.



- Line up the base line of the protractor over PM with the centre mark on P.
- Follow the inside scale around on the protractor, from 0° to 76°.
- Mark this point.



- Draw an interval from P through this mark and label the end of this interval K.



- This creates $\angle KPM$, measuring 76°.

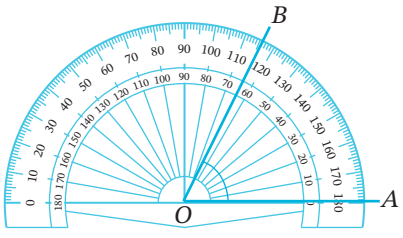
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EXERCISE 2.02 ANSWERS ON P. 564

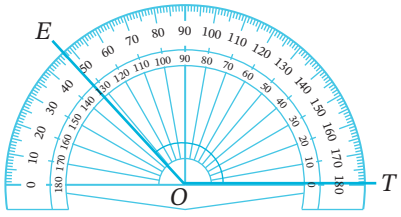
Measuring and drawing angles UFRC

1 Find the size of each marked angle.

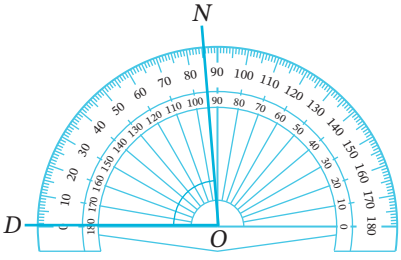
a



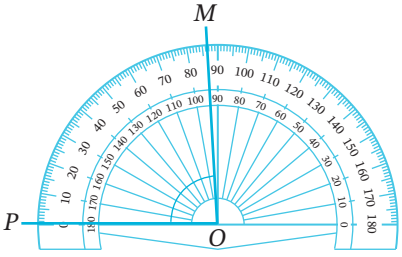
b



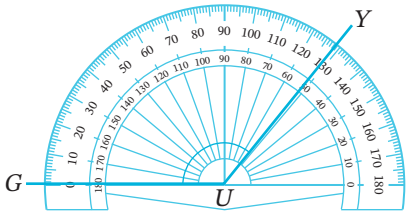
c



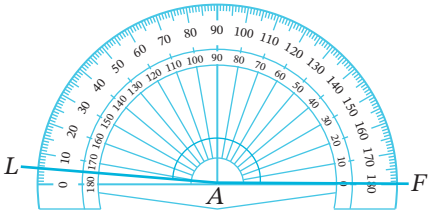
d



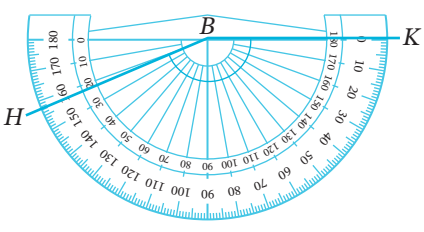
e



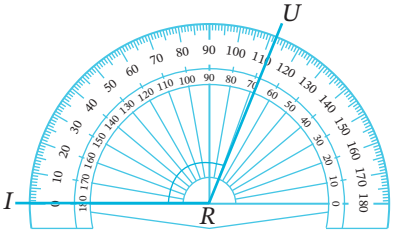
f



g



h

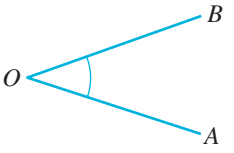


Foundation | Standard | Complex

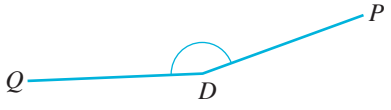
EXAMPLE 3

2 Estimate the size of each angle, then check your estimate by measuring with a protractor.

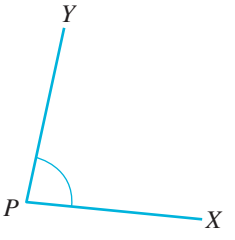
a



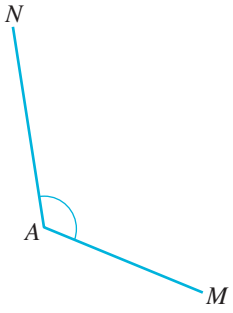
b



c



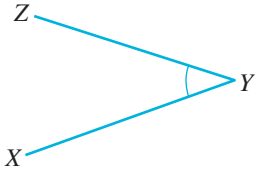
d



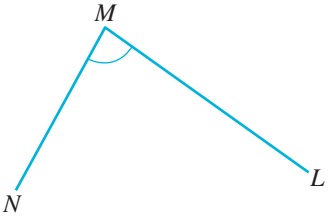
e



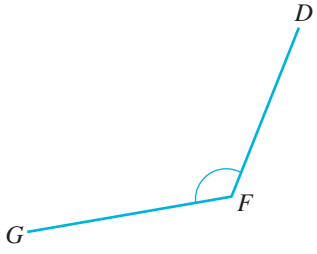
f



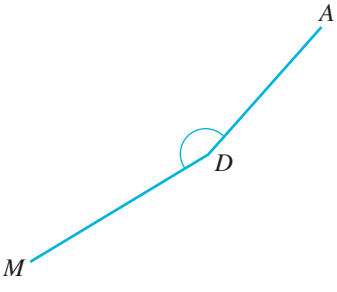
g



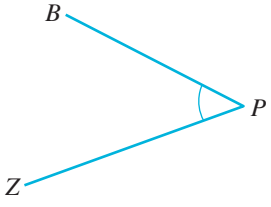
h



i



j

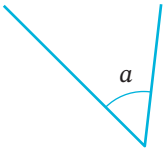


Foundation Standard Complex

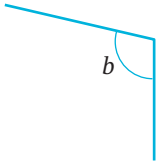
9780170477963

3 Sort the following angles from smallest to largest. R

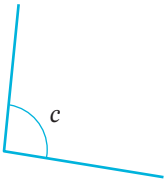
a



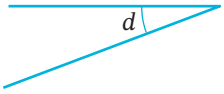
b



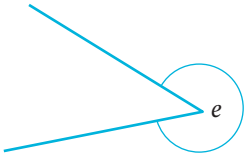
c



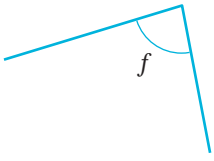
d



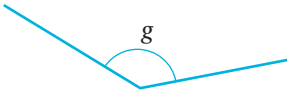
e



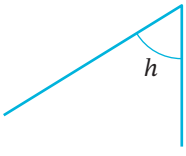
f



g



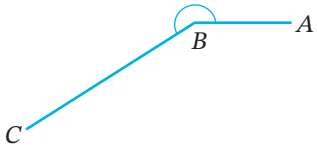
h



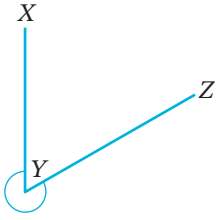
4 Estimate the size of each angle, then check by measuring with a protractor. R

EXAMPLE 4

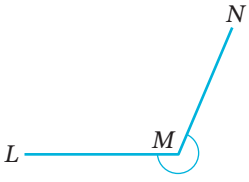
a



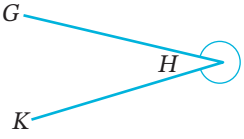
b



c



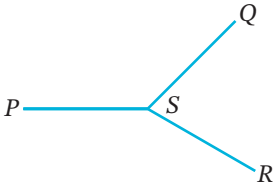
d



5 Measure the size of $\angle QSR$. Select **A**, **B**, **C** or **D**.

A 135° **B** 105°

C 75° **D** 70°



Foundation Standard Complex

6 This photo shows Daniel shooting for goal in a game of football. His shooting angle is shown on the photo. Estimate the size of this angle.
Select **A**, **B**, **C** or **D**. R


iStock.com/mikkelwilliam

A 60° **B** 80°
C 100° **D** 120°

7 The word 'degree' has many meanings. Find 4 non-mathematical meanings for 'degree'. C

EXAMPLE 5

8 Use a protractor to accurately construct an angle of size:

a 35° **b** 115° **c** 150° **d** 63°
e 17° **f** 170° **g** 204° **h** 319°

9 Use a protractor to accurately draw and label each angle.

a $\angle DRE = 65°$ **b** $\angle BGH = 145°$ **c** $\angle GRT = 32°$
d $\angle ABC = 45°$ **e** $\angle SAQ = 110°$ **f** $\angle NMH = 265°$
g $\angle KLY = 28°$ **h** $\angle LMN = 180°$ **i** $\angle LKY = 99°$

Did you know?

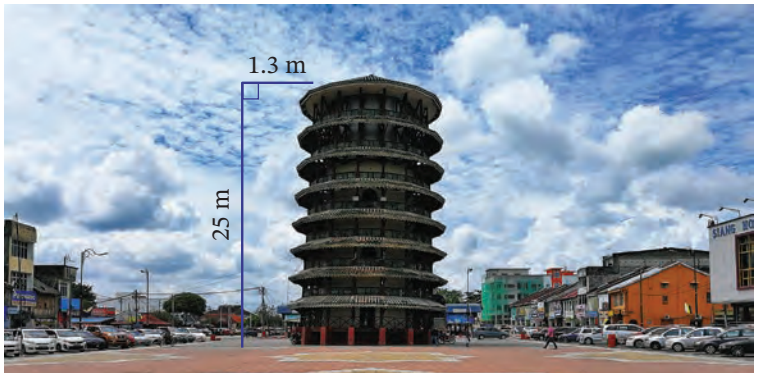
The Leaning Tower of Teluk Intan

The Leaning Tower of Pisa is not the only tower that leans.

The Clock Tower of Teluk Intan in Malaysia was built in 1885 as a water tower but became a watch tower during World War II. Although from the outside it looks like an 8 storey building, on the inside it has only 3 storeys.

The tower is 25 metres high and began leaning due to the soft soil foundation and the weight of the water tank inside. It now has a tilt of 3° from the vertical.

Draw a scale diagram of this tower. Research what the inside of the building looks like.



Shutterstock.com/fiz_ ero

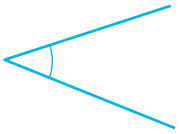
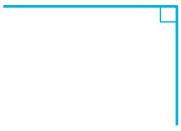
Foundation Standard Complex

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Types of angles

2.03

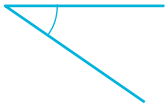
Angles are classified into types according to their size.

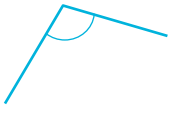
Angle	Type	Description
	**acute**	less than 90°
	**right**	90° (quarter turn) Note that a right angle is marked with a box symbol.
	**obtuse**	greater than 90° but less than 180°
	**straight**	180° (half turn)
	**reflex**	greater than 180° but less than 360°
	**revolution**	360° (complete turn)

EXERCISE 2.03 ANSWERS ON P. 564

Types of angles U F R C

1 State whether each angle is acute (A), obtuse (O) or reflex (R). R C

a 

b 

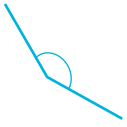
c 

d 

e 

f 

g 

h 

i 

Foundation Standard Complex

j



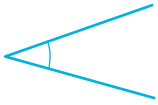
k



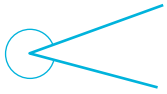
l



m



n



o



2 Draw 2 different:

a acute angles **b** obtuse angles **c** right angles

d reflex angles **e** straight angles **f** revolutions

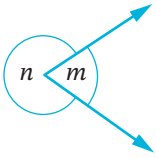
3 Classify each angle into one of the 6 types. R C

a	37°	**b**	107°	**c**	252°	**d**	195°	**e**	79°
f	180°	**g**	163°	**h**	179°	**i**	360°	**j**	5°
k	345°	**l**	91°	**m**	14°	**n**	299°	**o**	90°
p	205°	**q**	126°	**r**	44°				

4 What types of angles are m and n, respectively? Select **A**, **B**, **C** or **D**. R C

A obtuse and reflex **B** reflex and a revolution

C acute and obtuse **D** acute and reflex



5 Decide whether each marked angle is acute, obtuse or reflex. R C

a



iStock.com/Fitzer

b



Alamy Stock Photo/Bill Gozansky

c



Shutterstock.com/Real Window Creative

d



Shutterstock.com/Paul Drabot

Technology

Constructing angles 1

Using dynamic geometry software, construct and label angles of various sizes.

1 Draw, label and measure:

a an acute angle **b** a right angle

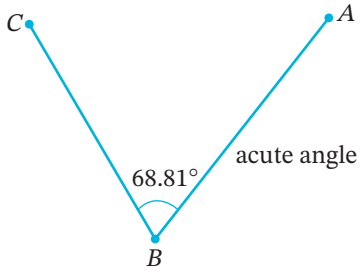
c an obtuse angle **d** a reflex angle

2 Start a new sketch and accurately construct separate angles of size:

a 72° **b** 310° **c** 165°

d 98° **e** 236° **f** 90°

3 Using **Insert Text**, classify each angle according to its type, for example, acute, reflex.

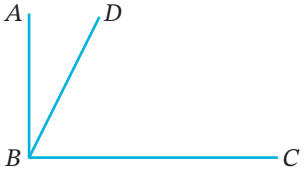


Investigation

Complementary and supplementary angles

1 For each diagram, measure the angles named, then copy and complete the equations.

a

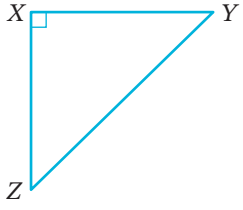


$\angle ABD =$ _____

$\angle CBD =$ _____

$\angle ABD + \angle CBD =$ _____

b



$\angle Y =$ _____

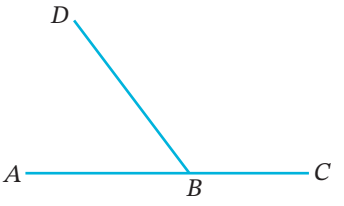
$\angle Z =$ _____

$\angle Y + \angle Z =$ _____

2 The 2 angles you measured in each diagram of question **1** are called **complementary angles**. What do complementary angles add up to?

3 For each diagram, measure the angles named, then copy and complete the equations.

a

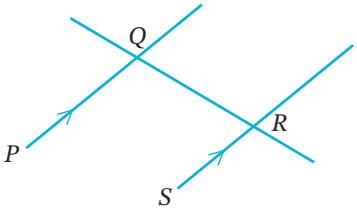


$\angle ABD =$ _____

$\angle CBD =$ _____

$\angle ABD + \angle CBD =$ _____

b



$\angle PQR =$ _____

$\angle SRQ =$ _____

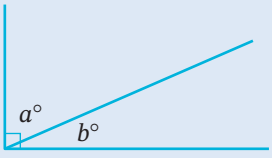
$\angle PQR + \angle SRQ =$ _____

4 The 2 angles you measured in each diagram of question **3** are called **supplementary angles**. What do supplementary angles add up to?

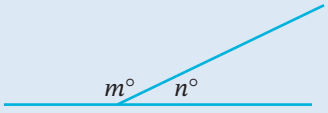
2.04 Complementary and supplementary angles

Complementary and supplementary angles

- **Complementary angles** add to 90°
- **Supplementary angles** add to 180°
- **Angles in a right angle** are complementary (add to up 90°)
- **Angles on a straight line** are supplementary (add to 180°)



$a + b = 90$

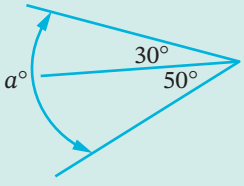


$m + n = 180$

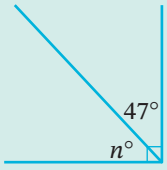
Example 6

In each diagram, there is an angle whose size is represented by a letter called a **variable** or **pronumeral**. A variable stands for a number. Find the value of each variable.

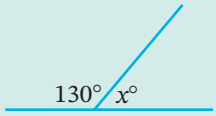
a



b



c



9780170477963

Solution

a $a = 30 + 50$ — Adding 2 adjacent angles.
$= 80$

b $n + 47 = 90$ — Angles in a right angle are complementary.
$n = 90 - 47$
$= 43$

c $x + 130 = 180$ — Angles on a straight line are supplementary.
$x = 180 - 130$
$= 50$

EXERCISE 2.04 ANSWERS ON P. 564

Complementary and supplementary angles U F R C

1 **a** Why is 57° the complement of 33°?
b Why is 147° the supplement of 33°? R C

2 Find the complement of:

a 30°	**b** 70°	**c** 25°	**d** 38°
e 89°	**f** 57°	**g** 42°	**h** 66°

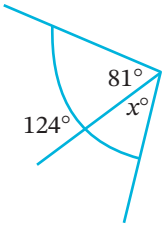
3 Look up 'complement' and 'supplement' in a dictionary. Find one non-mathematical meaning of each word. C

4 Find the supplement of:

a 18°	**b** 150°	**c** 35°	**d** 125°
e 62°	**f** 87°	**g** 111°	**h** 4°

5 Find the value of x. Select the correct answer **A**, **B**, **C** or **D**. R

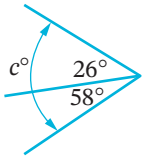
EXAMPLE 6



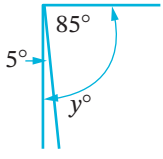
A 9 **B** 43
C 56 **D** 205

6 Find the value of the variable in each diagram. R

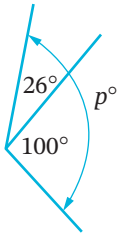
a



b

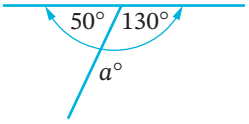


c

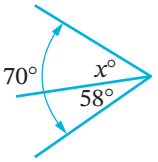


Foundation Standard Complex

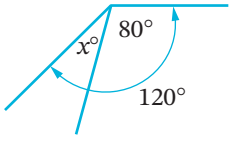
d



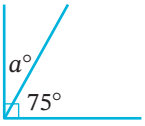
e



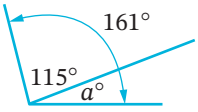
f



g



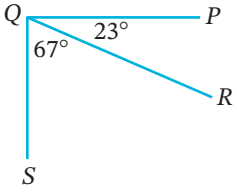
h



7 **a** Name the pair of complementary angles in this diagram.

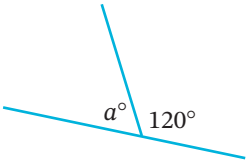
b How do you know that the angles you named are complementary?

c What type of angle is $\angle PQS$? R C

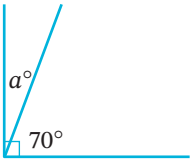


8 Find the value of the each variable. R

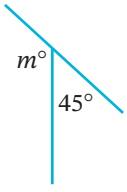
a



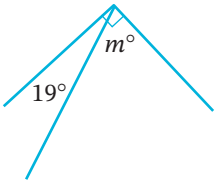
b



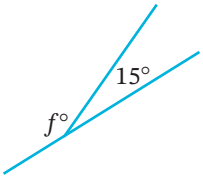
c



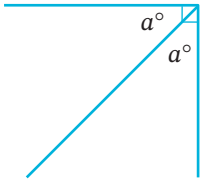
d



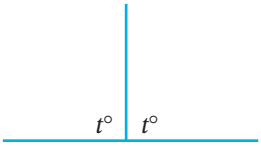
e



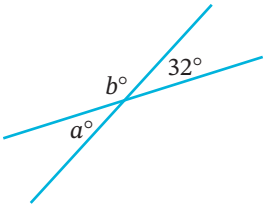
f



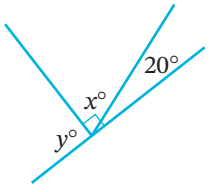
g



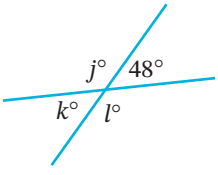
h



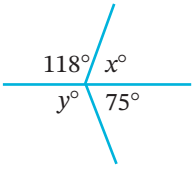
i



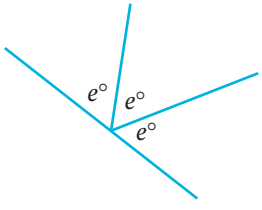
j



k



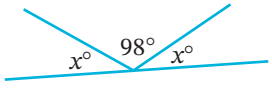
l



9 Find the value of x. Select **A**, **B**, **C** or **D**. R

A 41 **B** 42

C 82 **D** 131



Foundation Standard Complex

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Mental skills 2: Maths without calculators ANSWERS ON P. 564

Quiz
Mental skills 2

Adding or multiplying in any order

Have you noticed that $4 + 7 = 7 + 4$?
Have you noticed that $3 \times 5 = 5 \times 3$?
Numbers can be added or multiplied in any order. We can use this property to make our calculations simpler.

1 Study each example.

a $19 + 5 + 5 + 1 = (19 + 1) + (5 + 5)$
$= 20 + 10$
$= 30$

b $13 + 8 + 20 + 27 + 80 = (13 + 27) + (20 + 80) + 8$
$= 40 + 100 + 8$
$= 148$

c $2 \times 36 \times 5 = (2 \times 5) \times 36$
$= 10 \times 36$
$= 360$

d $25 \times 11 \times 4 \times 7 = (25 \times 4) \times (11 \times 7)$
$= 100 \times 77$
$= 7700$

2 Now evaluate each sum.

a $45 + 16 + 45 + 4 + 7$
b $38 + 600 + 50 + 12 + 40$
c $18 + 91 + 9 + 20$
d $75 + 33 + 7 + 25$
e $24 + 16 + 80 + 44 + 10$
f $56 + 5 + 20 + 15 + 4$
g $100 + 36 + 200 + 10 + 90$
h $54 + 27 + 9 + 16 + 3$
i $70 + 50 + 30 + 25 + 25$
j $32 + 120 + 40 + 80 + 40$

3 Now evaluate each product.

a $8 \times 4 \times 5$
b $50 \times 7 \times 2$
c $3 \times 5 \times 6$
d $5 \times 11 \times 40$
e $12 \times 2 \times 3$
f $2 \times 4 \times 25 \times 8$
g $3 \times 20 \times 7 \times 5$
h $6 \times 8 \times 5 \times 2$
i $2 \times 3 \times 2 \times 11$

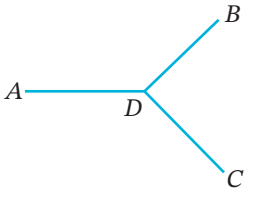
Investigation

Angles at a point and vertically opposite angles

1 In each diagram, there are adjacent angles that meet at a central point.

Measure all angles, then copy and complete the equations.

a



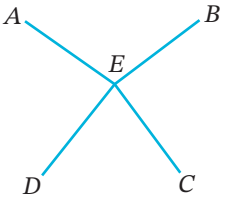
$\angle ADB =$ _____

$\angle ADC =$ _____

$\angle BDC =$ _____

$\angle ADB + \angle ADC + \angle BDC =$ _____

b



$\angle AEB =$ _____

$\angle BEC =$ _____

$\angle CED =$ _____

$\angle DEA =$ _____

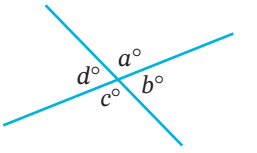
$\angle AEB + \angle BEC + \angle CED + \angle DEA =$ _____

2 What do the angles that meet at a point add up to? Why?

3 When 2 lines cross, 4 angles are created.

a Which of these angles are equal?

b Can you prove it using supplementary angles?



2.05 Angles at a point and vertically opposite angles

Puzzles
Angles: a dog day

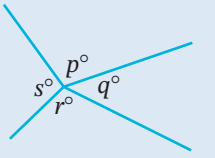
Angles

Worksheet
Straight angles, right angles and revolutions

Angles at a point

Angles at a point (in a revolution) add to 360°

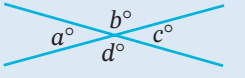
$p + q + r + s = 360$



Vertically opposite angles

Vertically opposite angles are equal.

$a = c$ and $b = d$

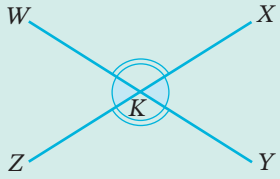


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Example 7

Name the angle that is vertically opposite:

a $\angle WKZ$ **b** $\angle ZKY$



Solution

a $\angle XKY$ is vertically opposite $\angle WKZ$

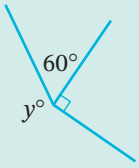
b $\angle WKX$ is vertically opposite $\angle ZKY$

Video
Angles at a point and vertically opposite angles

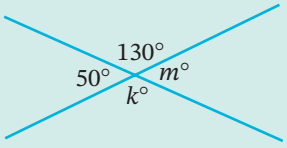
Example 8

Find the value of the variable in each diagram.

a



b



Solution

a $y + 60 + 90 = 360$ Angles at a point add to 360°

$y = 360 - 60 - 90$

$= 210$

b $k = 130$ Vertically opposite angles are equal

$m = 50$ Vertically opposite angles are equal

Videos
Angles at a point and vertically opposite angles
Angle geometry

2.05

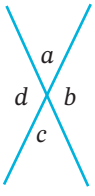
EXERCISE 2.05 ANSWERS ON P. 564

Angles at a point and vertically opposite angles UFRC

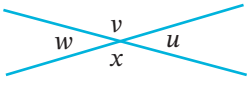
EXAMPLE 7

1 Write the angle that is vertically opposite each named angle. R C

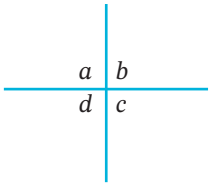
a a



b w

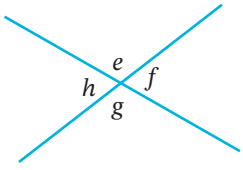


c c

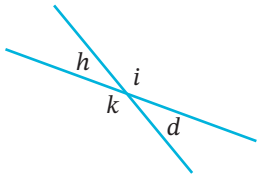


Foundation Standard Complex

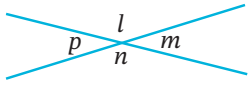
d h



e k



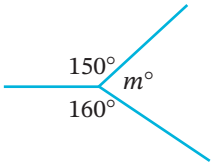
f m



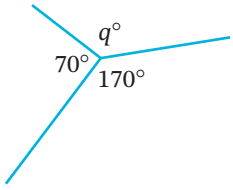
EXAMPLE 8

2 Find the value of each variable. R

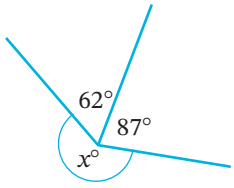
a



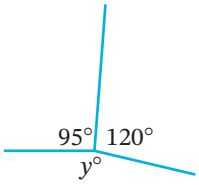
b



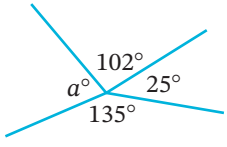
c



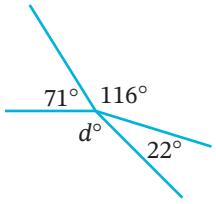
d



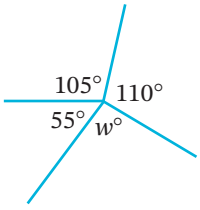
e



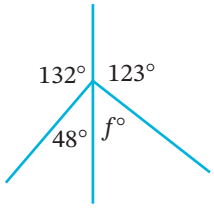
f



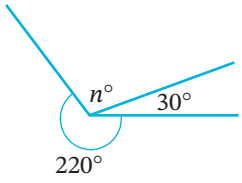
g



h



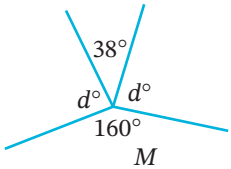
i



3 Find the value of d. Select the correct answer **A**, **B**, **C** or **D**. R

A 122 **B** 61

C 142 **D** 81

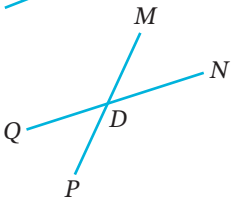


4 **a** Which angle is vertically opposite $\angle NDP$?

b Which angle is equal to $\angle MDQ$?

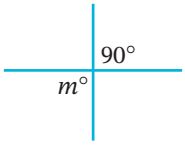
c Name 2 straight angles.

d Name 2 different pairs of supplementary angles. R C

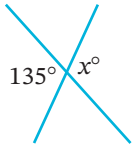


5 Find the value of each variable. C

a



b



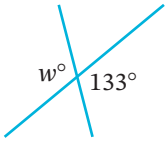
c



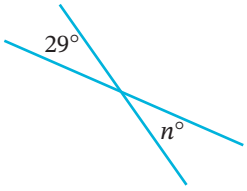
Foundation Standard Complex

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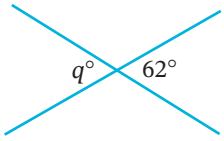
d



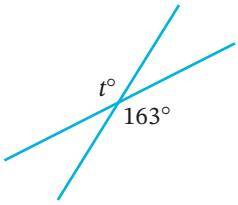
e



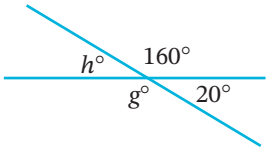
f



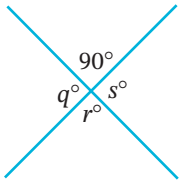
g



h

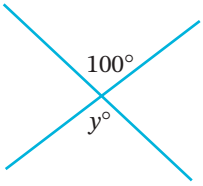


i

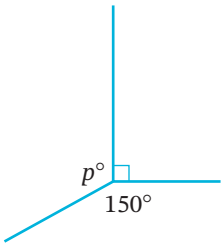


6 Find the value of each variable. R

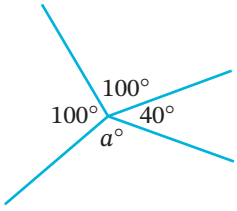
a



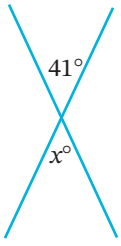
b



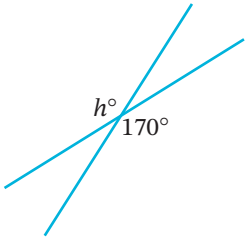
c



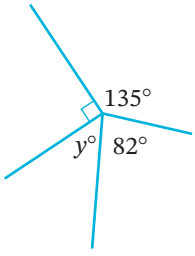
d



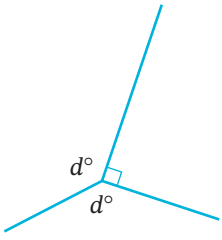
e



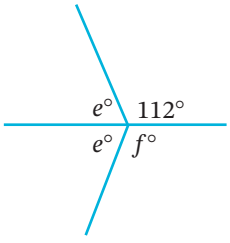
f



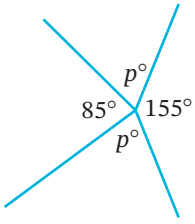
g



h



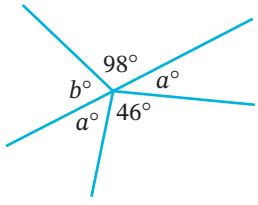
i



7 Find the value of a and b. Select **A**, **B**, **C** or **D**. R

A $a = 46, b = 82$ **B** $a = 54, b = 80$

C $a = 67, b = 46$ **D** $a = 67, b = 82$



Foundation Standard Complex

Did you know?

Why 360 degrees?



Shutterstock.com/Ali Ender Birer

Why are there 90° in a right angle and 360° in a revolution? Why do we use such unusual numbers instead of more conventional numbers like 10 and 100?

Over 4000 years ago, in 2000 BCE, the ancient Babylonians used a base 60 system of numbers. They used a base 60 number system because 60 has more factors than 10. You can divide 60 by 2, 3, 4, 5, 6, 10, 12, 15, 20 and 30.

Furthermore, $6 \times 60 = 360$, which was the Babylonian approximation of the number of days in a year. They defined a revolution as being 360° so that each day the Earth would travel 1° around the Sun. A right angle, being a quarter-revolution, thus became $360° \div 4 = 90°$.

Some people who prefer a base 10 system of measuring angles use **grads** instead of degrees. With this system, a right angle is 100 grads and a revolution is 400 grads.

Find out more information about grads, including the exact relationship between degrees and grads.

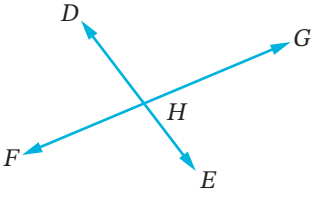
2.06 Parallel and perpendicular lines

Worksheet
A page of intervals

A line is named using 2 points on the line. For example, this is the line *AB*.



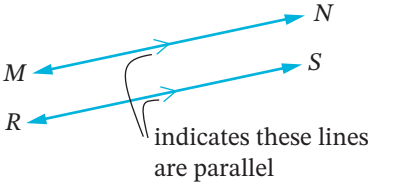
When 2 lines cross, we say that they **intersect** at a point. For example, in this diagram, line *DE* intersects line *FG* at the point *H*.



Parallel lines

Lines that point in the same direction and never intersect are called **parallel lines**. Parallel lines are marked with identical arrowheads and are always the same distance apart. For example, in this diagram, *MN* is parallel to *RS*.

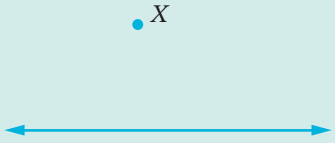
This is written as '*MN* || *RS*', where the symbol || stands for 'is parallel to'.



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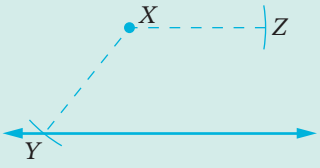
Example 9

Use compasses to construct a line through X that is parallel to the given line.



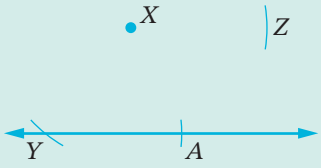
Solution

Step 1



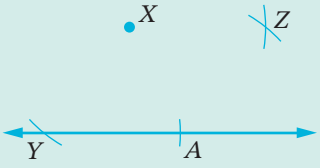
Use compasses from X to mark 2 large arcs at Y and Z.

Step 2



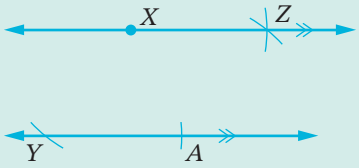
Use compasses from Y to mark an arc with the same radius at A on the line.

Step 3



Use compasses from A to mark an arc with the same radius to cross the arc at Z.

Step 4

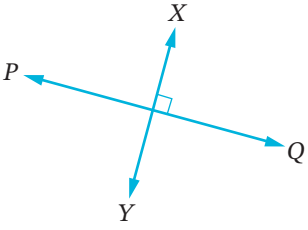


Join XZ to construct a line parallel to AY.

Perpendicular lines

Lines that intersect at right angles (90°) are called **perpendicular lines**. For example, in this diagram, PQ is perpendicular to XY.

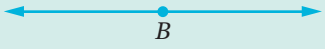
This is written as '$PQ \perp XY$', where the $\perp$ symbol stands for 'is perpendicular to'.



EXTENSION

Example 10

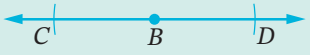
Use compasses to construct a perpendicular through point B on the line.



Solution

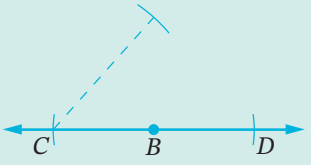
Constructing a perpendicular to a point on a line is the same as bisecting a straight angle (180°).

Step 1



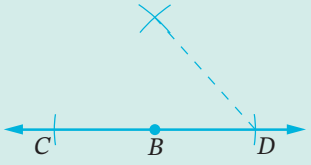
Use compasses to draw 2 arcs from B.

Step 2



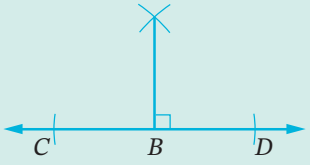
Open compasses wider to draw an arc from C.

Step 3



Use compasses to draw an arc with the same distance from D.

Step 4

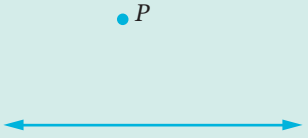


Join B to where the 2 arcs cross.

Use a protractor or set square to check that the line is perpendicular (at 90°) to CD.

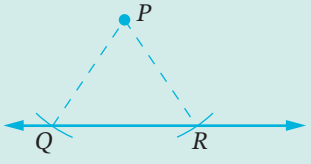
Example 11

Use compasses to draw a perpendicular through point P above the line.



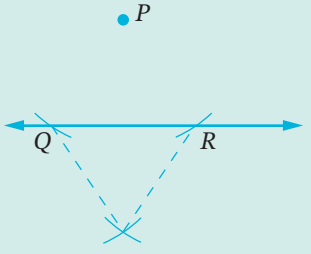
Solution

Step 1



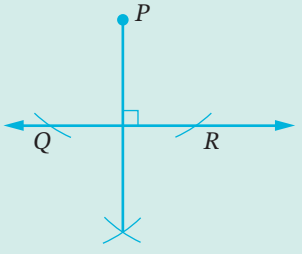
Use compasses from P to mark 2 arcs with the same radius on the line.

Step 2



Use compasses from Q and R to mark 2 intersecting arcs with the same radius below the line.

Step 3



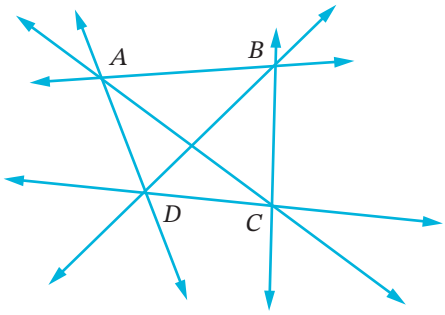
Join P to where the 2 arcs cross.

Use a protractor or set square to check that the line is perpendicular to QR.

EXERCISE 2.06 ANSWERS ON P. 565

Parallel and perpendicular lines U F R C

1 Name the 6 different lines in this diagram. C

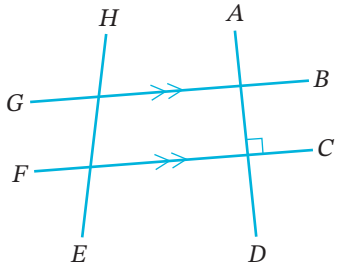


2 In this diagram, name 2 lines that: C

a are perpendicular

b are parallel

c intersect



3 Rewrite your answers to question **2** parts **a** and **b** using the symbols for 'is perpendicular to' and 'is parallel to'. C

4 Draw and label correctly: C

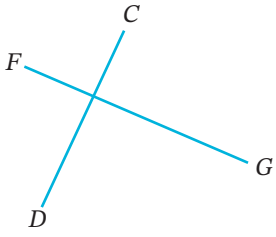
a line FG

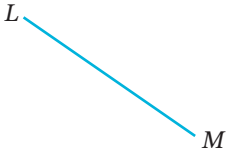
b line AB intersecting line CD at point E

c line PQ parallel to line YZ

d line JK perpendicular to line LM.

5 An **interval** is a section of a line, with a definite length. Which interval is parallel to FG? Select the correct answer **A**, **B**, **C** or **D**.









A CD **B** LM **C** AB **D** PQ

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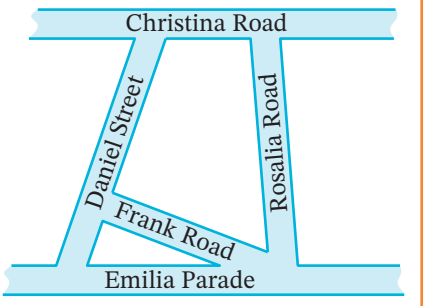
6 On the map, what is perpendicular to Frank Road? Select **A**, **B**, **C** or **D**. C

A Emilia Parade

B Rosalia Road

C Daniel Street

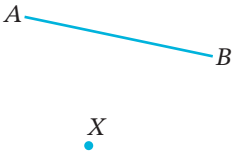
D Christina Road



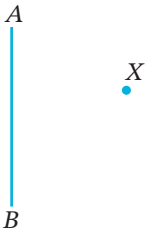
EXAMPLE 9

7 Copy each diagram and construct or draw a line parallel to AB through X.

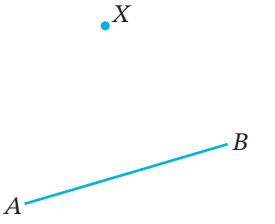
a



b



c



EXAMPLE 10

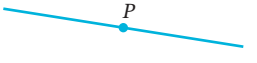
8 Draw a line and mark a point, L, on it. Construct a perpendicular line through L using compasses or a protractor.

EXAMPLE 10

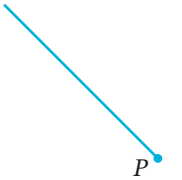
9 Draw a line and mark a point, X, above it. Construct or draw a perpendicular to the line that passes through X.

10 Copy each diagram and construct or draw a perpendicular line through P.

a



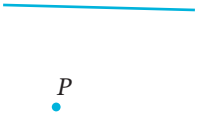
b



c



d



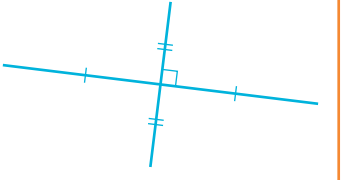
11a Draw 2 different-sized intervals that are parallel.

b Join their ends to make a quadrilateral.

c What type of quadrilateral have you constructed? R C

12a Draw an interval and mark its midpoint.

b Draw a different-sized interval through the midpoint of the first interval, perpendicular to it and with the same midpoint (as shown).



c Join the ends of both intervals to make a quadrilateral.

d What type of quadrilateral have you constructed? R C

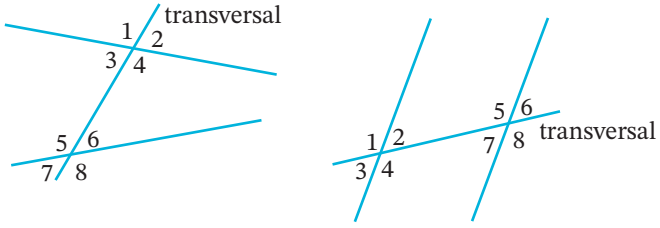
Foundation Standard Complex

9780170477963

Corresponding angles on parallel lines 2.07

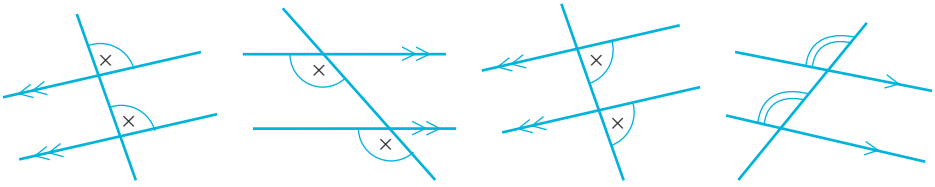
A line that crosses 2 or more other lines is called a **transversal**. Transverse means 'crossing'.

If a transversal crosses 2 lines, then 8 angles are created.



These 8 angles have special properties if the transversal crosses *parallel* lines.

Corresponding angles on parallel lines are any 2 angles in matching positions according to the transversal and a parallel line, as shown below.



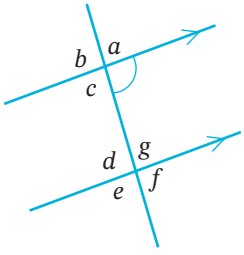
'Corresponding' means 'matching', and corresponding angles on parallel lines are equal in size.

EXERCISE 2.07 ANSWERS ON P. 565

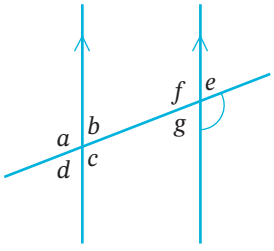
Corresponding angles on parallel lines U F R C

1 Write the corresponding angle to each marked angle. R C

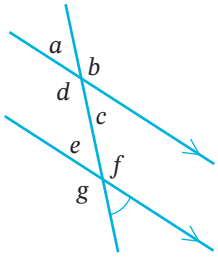
a



b

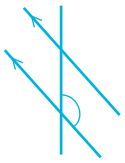


c

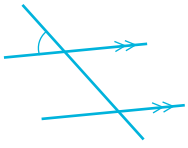


2 Copy each diagram and mark the corresponding angle to the marked angle. R

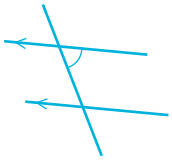
a



b



c



Worksheet
Investigating angles on parallel lines

Skillsheet
Angles and parallel lines

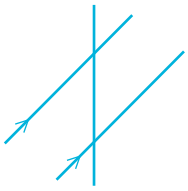
Video
Angle relationships

2.07

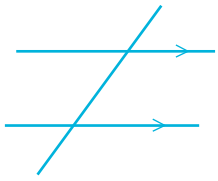
Foundation | Standard | Complex

3 Copy each diagram and mark any pair of corresponding angles. R

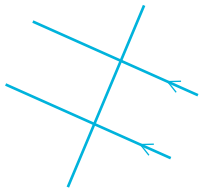
a



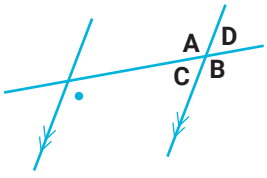
b



c

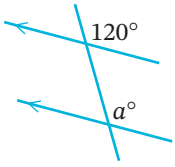


4 Which angle is corresponding to the angle marked • ?
Select the correct answer **A**, **B**, **C** or **D**. R

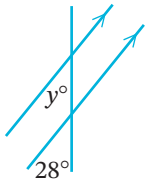


5 Find the value of each variable. R

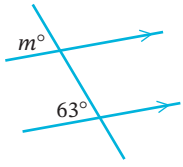
a



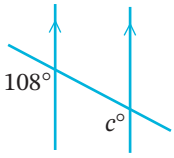
b



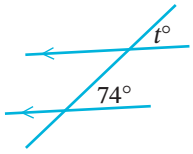
c



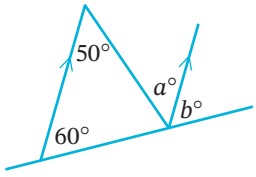
d



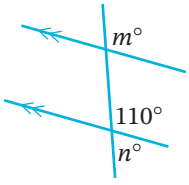
e



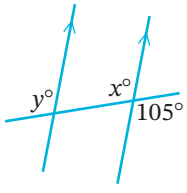
f



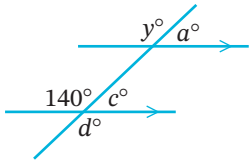
g



h

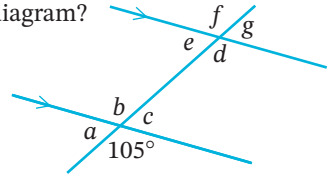


i



6 **a** How many pairs of corresponding angles are in this diagram?

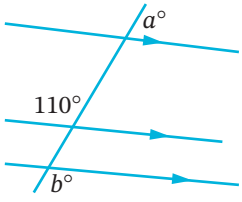
b Find the sizes of angles a to g. R



7 Find the values of a and b. Select **A**, **B**, **C** or **D**. R

A $a = 110, b = 110$ **B** $a = 70, b = 70$

C $a = 70, b = 110$ **D** $a = 110, b = 70$



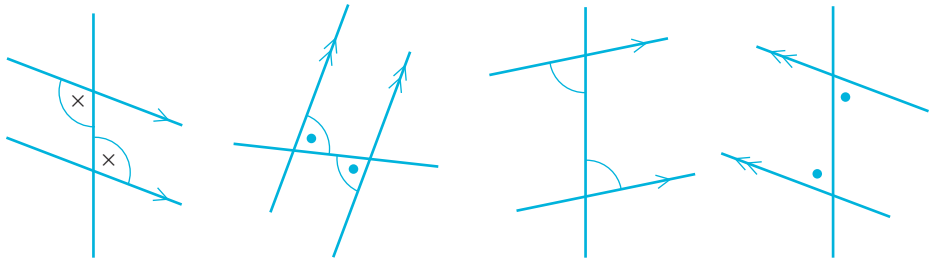
Foundation Standard Complex

9780170477963

Alternate angles on parallel lines

2.08

Alternate angles on parallel lines are any 2 angles *between* the lines and on *opposite* sides of the transversal, as shown below.



'Alternate' means 'changing direction'.

Worksheet Investigating angles on parallel lines

Skillsheet Angles and parallel lines

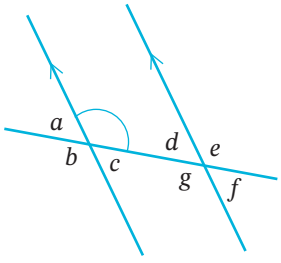
Video Angle relationships

EXERCISE 2.08 ANSWERS ON P. 565

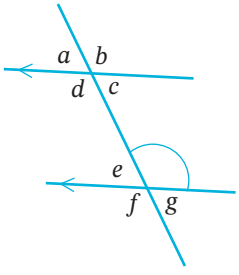
Alternate angles on parallel lines U F R C

1 Write the alternate angle to each marked angle. R C

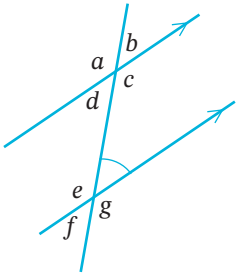
a



b

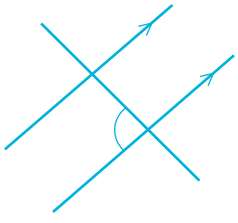


c

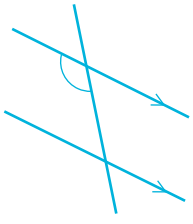


2 Copy each diagram and mark the alternate angle to the marked angle. R

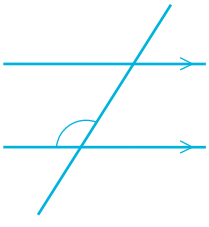
a



b

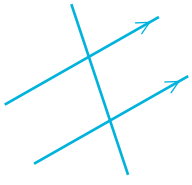


c

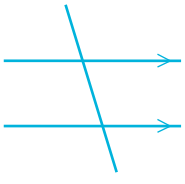


3 Copy each diagram and mark any pair of alternate angles. R

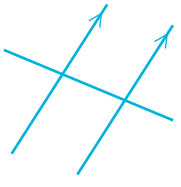
a



b



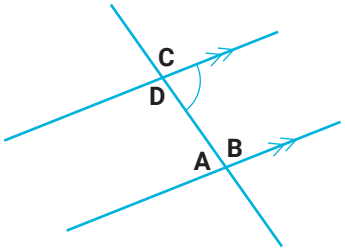
c



Foundation Standard Complex

9780170477963

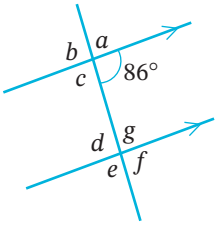
4 Which angle is alternate to the marked angle?
Select the correct answer **A**, **B**, **C** or **D**. R



5 **a** How many pairs of alternate angles are there in this diagram?

b Find the sizes of the angles a to g.

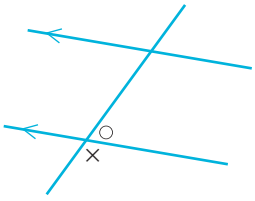
c What is the relationship between the alternate angles on parallel lines? R C



6 Two angles have been marked on this diagram. The other 6 angles are either ○ or ×.

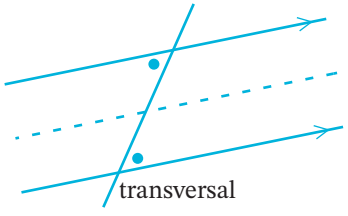
a Copy and complete the diagram with ○s or ×s.

b What do you notice about the pairs of alternate angles? R C



7 **a** Copy this diagram on a sheet of paper and spin it upside down to show that it has rotational symmetry. What does that mean about the alternate angles?

b Cut out your diagram and then cut along the dotted line. Spin one of the alternate angles to fit exactly on top of the other. Are they the same size? R C



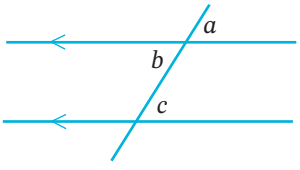
8 Examine this diagram.

a Why is $b = a$?

b Why is $c = a$?

c So what are the values of b and c?

d What does this prove about alternate angles on parallel lines? R C

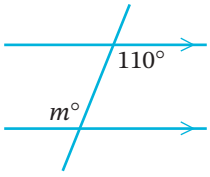


Foundation Standard Complex

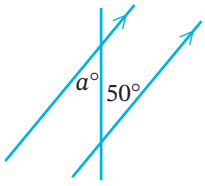
9780170477963

9 Find the value of each variable. R

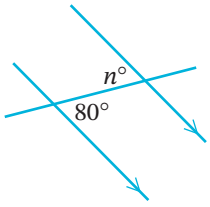
a



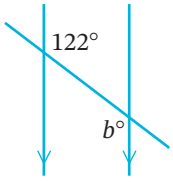
b



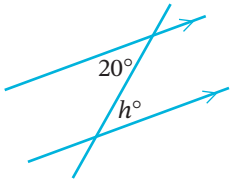
c



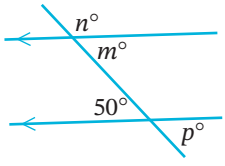
d



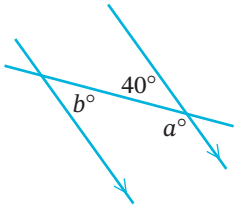
e



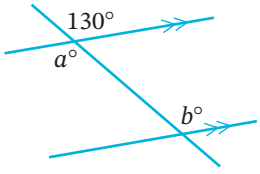
f



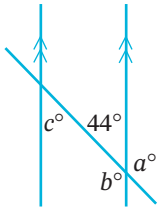
g



h



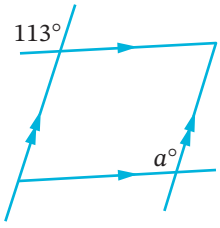
i



10 Find the value of a. Select **A**, **B**, **C** or **D**. R

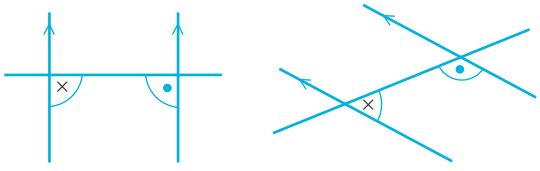
A 113 **B** 67

C 117 **D** 247



Co-interior angles on parallel lines 2.09

Co-interior angles on parallel lines are any 2 angles *between* the lines and on the *same* side of the transversal, as shown below.



'Co-interior' means 'together inside'.

Worksheets
Investigating angles on parallel lines

Angles in parallel lines

Skillsheet
Angles and parallel lines

Video
Angle relationships

Foundation Standard Complex

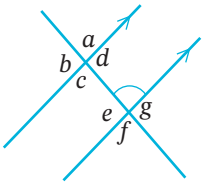
Puzzle
Angles in parallel lines

EXERCISE 2.09 ANSWERS ON P. 566

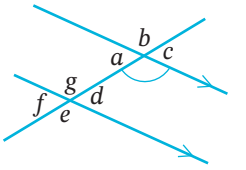
Co-interior angles on parallel lines U F R C

1 Write the co-interior angle to each marked angle. R C

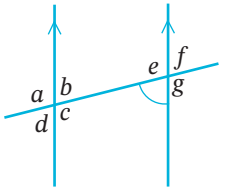
a



b

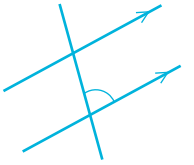


c

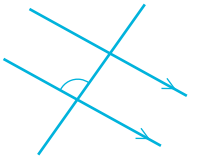


2 Copy each diagram and mark the co-interior angle to the marked angle. R

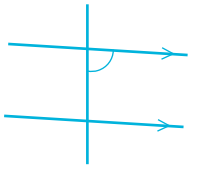
a



b

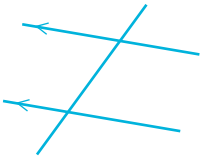


c

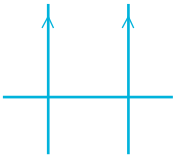


3 Copy each diagram and mark any pair of co-interior angles. R

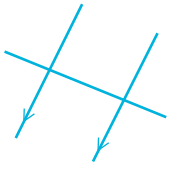
a



b



c

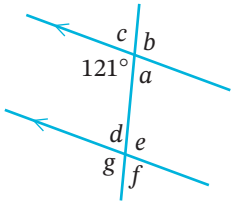


4 **a** How many pairs of co-interior angles are on this diagram?

b Which angle is co-interior to the 121° angle?
Select the correct answer **A**, **B**, **C** or **D**.

A d **B** b

C e **D** g

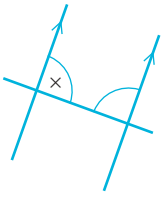


c Find the sizes of angles a to g.

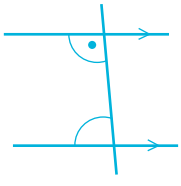
d What is the relationship between the co-interior angles on parallel lines? R C

5 Copy each diagram, measure the marked angles and calculate their sum.
What do you notice about your answer? C

a



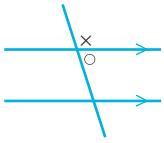
b



Foundation Standard Complex

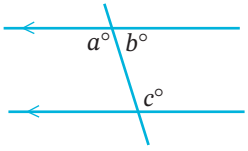
9780170477963

6 Two angles have been marked on this diagram.



- **a** How many degrees is ○ + ×? Why?
- **b** The other 6 angles are either ○ or ×. Copy and complete the diagram with ○s or ×s.
- **c** What do you notice about the pairs of co-interior angles? R C

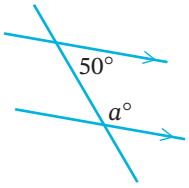
7 Examine this diagram.



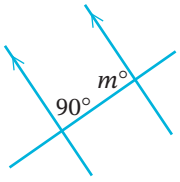
- **a** What is the value of $a + b$? Why?
- **b** Why is $a = c$?
- **c** So what is the value of $c + b$?
- **d** What does this prove about co-interior angles on parallel lines? R C

8 Find the value of each variable. C

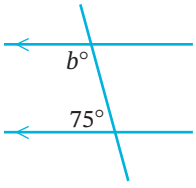
a



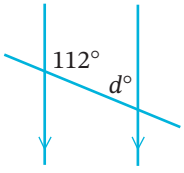
b



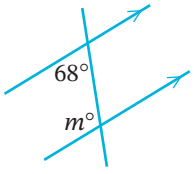
c



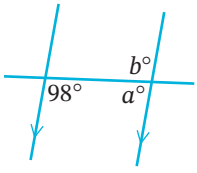
d



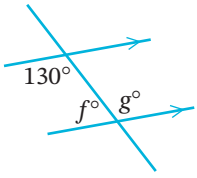
e



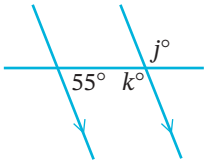
f



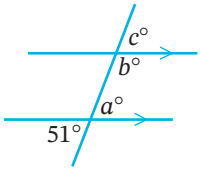
g



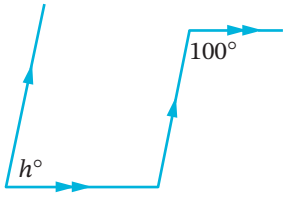
h



i



9 Find the value of h. Select **A**, **B**, **C** or **D**. R



A 20 **B** 80

C 100 **D** 260

2.10 Angles on parallel lines

Worksheets
Find the unknown angle

What is the diagram?

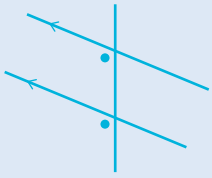
Matching angles

Skillsheet
Angles and parallel lines

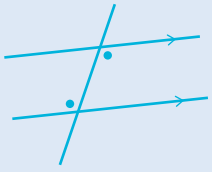
Angles on parallel lines

When parallel lines are crossed by a transversal:

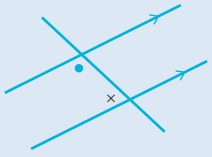
- corresponding angles are equal



- alternate angles are equal



- co-interior angles are supplementary (add to 180°).

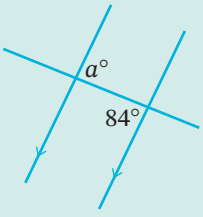


Video
Angles on parallel lines

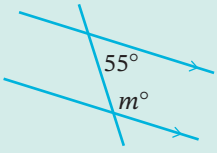
Example 12

Find the value of each variable, giving a reason.

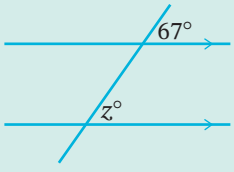
a



b



c



Solution

a $a = 84$ (Alternate angles on parallel lines)

b $m + 55 = 180$ (Co-interior angles on parallel lines)

$m = 180 - 55$

$= 125$

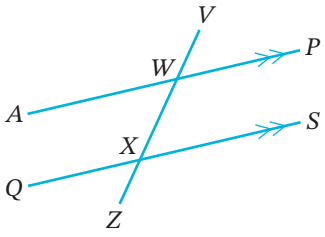
c $z = 67$ (Corresponding angles on parallel lines)

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EXERCISE 2.10 ANSWERS ON P. 566

Angles on parallel lines U F R C

1 For this diagram, name the angle that is: R C



a corresponding to $\angle VWA$

b alternate to $\angle QXW$

c co-interior with $\angle PWX$

d co-interior with $\angle AWX$

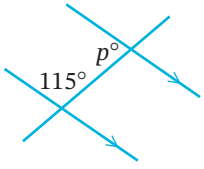
e alternate to $\angle SXV$

f corresponding to $\angle ZXS$.

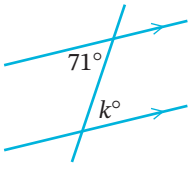
EXAMPLE 12

2 Find the value of each variable, giving a reason. R C

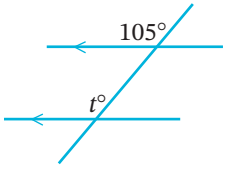
a



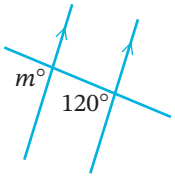
b



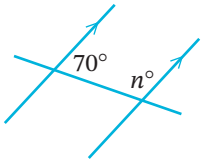
c



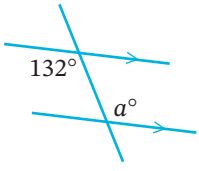
d



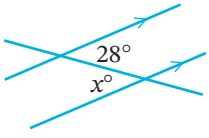
e



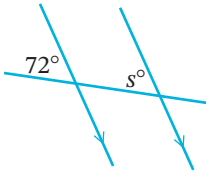
f



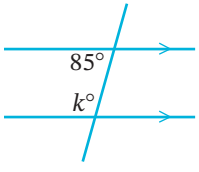
g



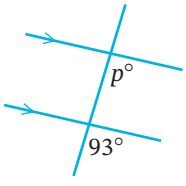
h



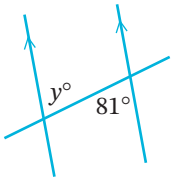
i



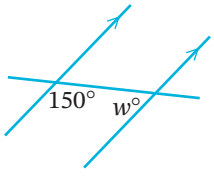
j



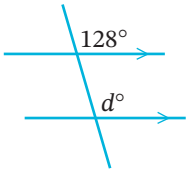
k



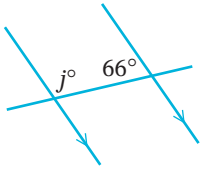
l



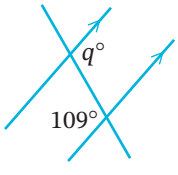
m



n



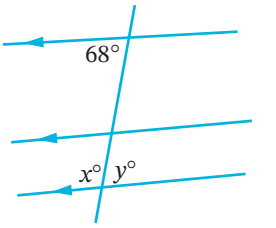
o



Foundation | Standard | Complex

9780170477963

3 Find the value of x and y.
Select the correct answer **A**, **B**, **C** or **D**. R

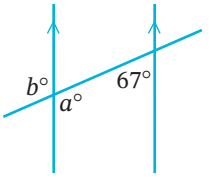


A $x = 22, y = 68$ **B** $x = 68, y = 112$

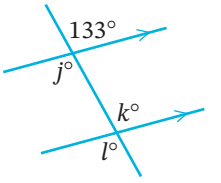
C $x = 112, y = 68$ **D** $x = 68, y = 22$

4 Find the value of each variable. R C

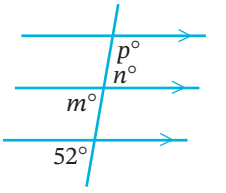
a



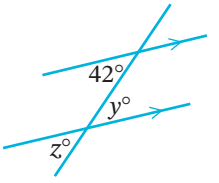
b



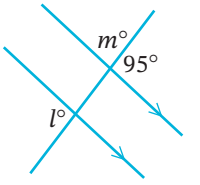
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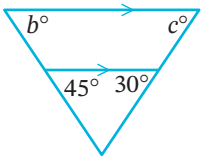
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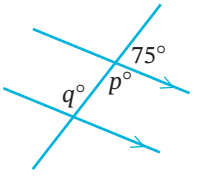
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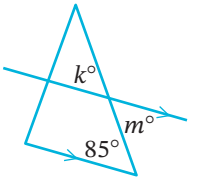
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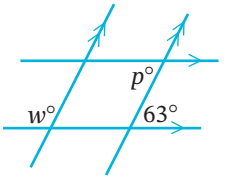
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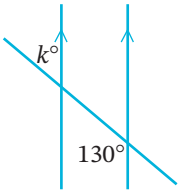
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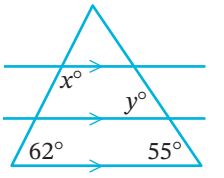
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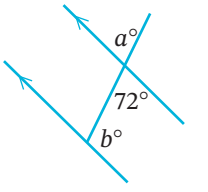
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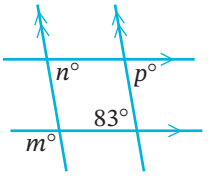
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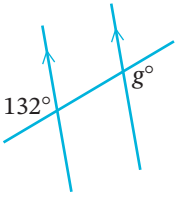
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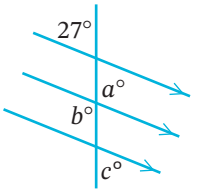
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o



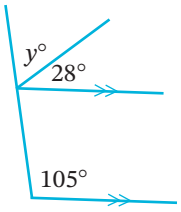
Foundation Standard Complex

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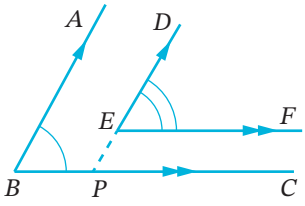
5 Which is the value of y? Select **A**, **B**, **C** or **D**. R

A 28 **B** 47

C 77 **D** 152



6 Prove that $\angle ABC = \angle DEF$. R C

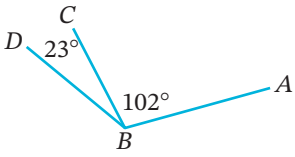


Technology

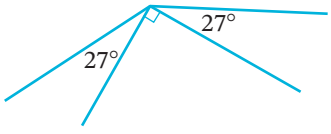
Constructing angles 2

1 Use dynamic geometry software to construct the following angles accurately.

a



b



c 2 complementary angles

d 2 supplementary angles

2 Construct each pair of angles on parallel lines.

a Corresponding angles of 28°

b Alternate angles of 65°

c Co-interior angles, where one of the angles is 130°

Proving parallel lines 2.11

We can use what we know about angles and parallel lines to prove that 2 lines are parallel.

Proving parallel lines

Two lines are parallel if:

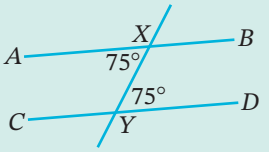
- alternate angles are equal, or
- corresponding angles are equal, or
- co-interior angles are supplementary (add up to 180°).

Foundation Standard Complex

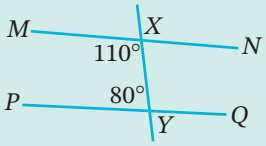
Example 13

Prove whether each pair of lines is parallel.

a $AB \parallel CD$



b $MN \parallel PQ$



Solution

a $\angle AXY$ is alternate to $\angle DYX$.

$\angle AXY = \angle DYX = 75°$

$\therefore AB \parallel CD$ (alternate angles are equal)

$\therefore$ means 'therefore'

b $\angle MXY$ is co-interior to $\angle PYX$

$\angle MXY + \angle PYX = 110° + 80°$

$= 190°$

$\neq 180°$

$\therefore MN$ is *not* parallel to PQ (co-interior angles *do not* add to 180°)

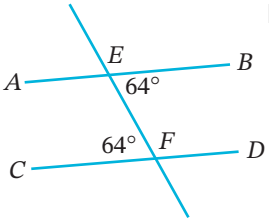
EXERCISE 2.11 ANSWERS ON P. 566

Proving parallel lines U F R C

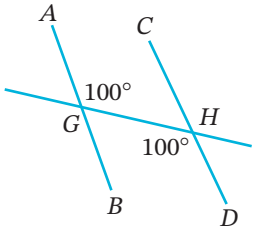
EXAMPLE 13

1 For each diagram, prove whether $AB \parallel CD$. R C

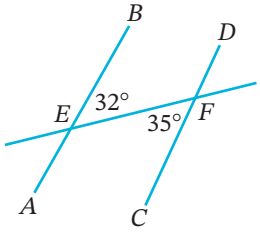
a



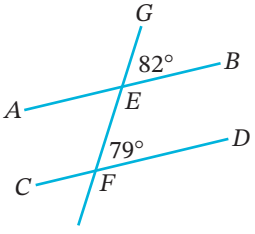
b



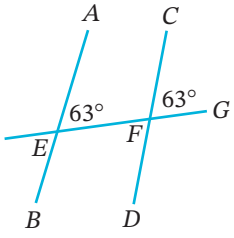
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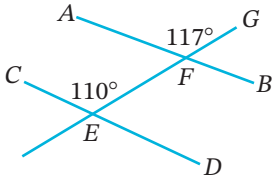
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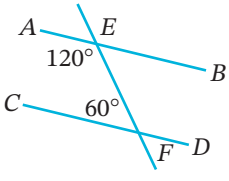
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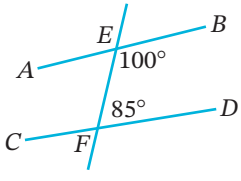
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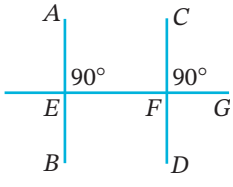
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i

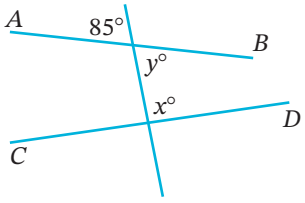


Foundation Standard Complex

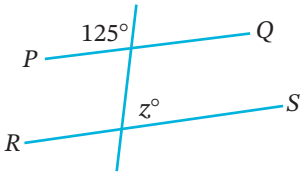
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2 **a** Find the value of y, giving a reason.

b Which value of x would prove that $AB \parallel CD$? R

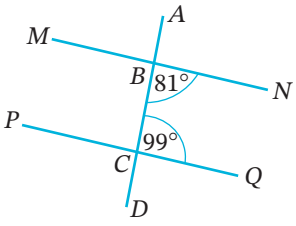


3 Which value of z would prove that $PQ \parallel RS$? R

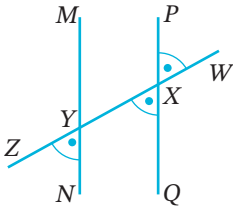


4 For each diagram, prove whether $PQ \parallel MN$, giving a reason. R C

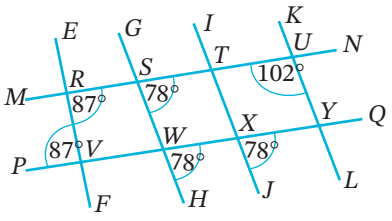
a



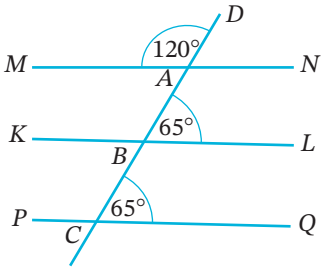
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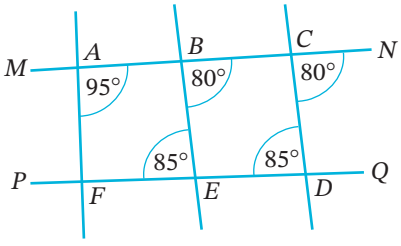
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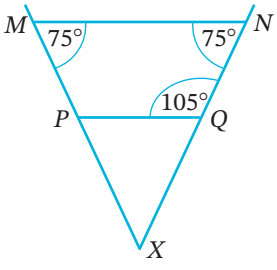
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f



Foundation | Standard | Complex

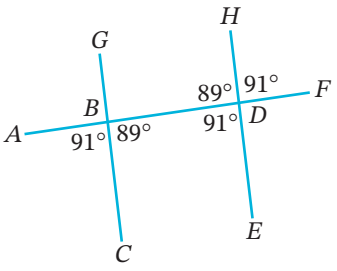
5 What reason proves that $GC \parallel HE$?
Select the correct answer **A**, **B**, **C** or **D**. R

A $\angle ABC = \angle HDF$ (alternate angles)

B $\angle CBD = \angle BDH$ (alternate angles)

C $\angle ADE = 91°$ (corresponding angles)

D $\angle BDE = \angle FDH$ (vertically opposite angles)



Power plus ANSWERS ON P. 567

1 How many degrees does the Earth spin on its axis in:

a one day? **b** one hour? **c** 8 hours? **d** 10 minutes?

2 Work out which direction (left, right, front or back) you would be facing after making each series of turns.

a Right 80°, right 240°, left 90°, right 40°

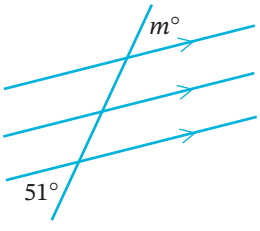
b Left 140°, left 140°, left 140°, right 60°

c Right 200°, left 70°, right 40°, right 10°

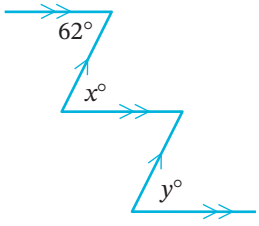
d Left 240°, right 190°, right 100°, left 50°

3 Find the value of each variable, giving reasons.

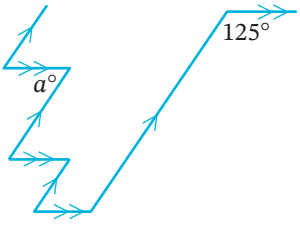
a



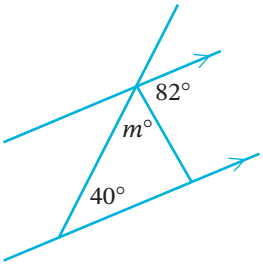
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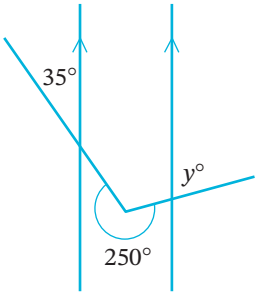
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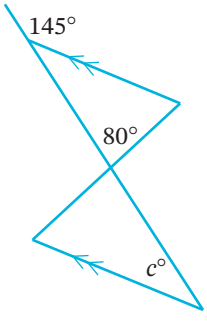
d



e



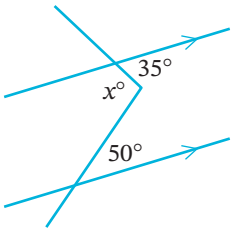
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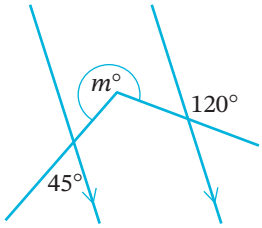
Foundation Standard Complex

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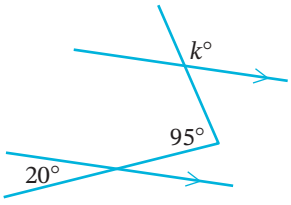
g



h



i



4 Draw an interval DE 6 cm in length. At D draw a line at any angle. On this line, mark a point G, also 6 cm from D. Construct a line through G parallel to DE. Construct a line through E parallel to DG. What shape have you drawn?

CHAPTER 2 REVIEW

Language of maths

Quiz Language of maths 2

Puzzle Angles crossword

acute	adjacent	alternate	angles at a point
co-interior	complementary	construct	corresponding
degrees	intersect	interval	obtuse
parallel	perpendicular	pronumeral	protractor
revolution	right	straight	supplementary
transversal	variable	vertex	vertically opposite

1 What is the name given to a line that crosses 2 or more other lines?

2 Find the meaning of 'acute' when referring to a disease, for example, acute appendicitis.

3 What is the difference between 'complementary' and 'complimentary'?

4 When something happens that dramatically changes the way we think or do things, it is said to be 'revolutionary'. Why do you think this word is used?

5 Draw 'angles at a point' and describe their property.

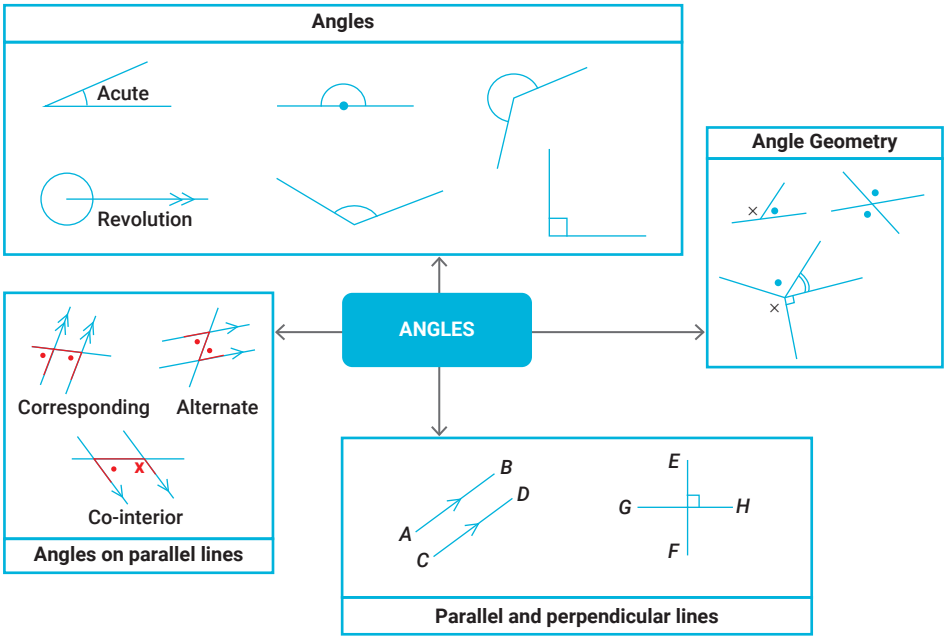
6 Mr Transversal visits his parents on **alternate** days. What does this mean? How is it similar to the mathematical meaning of 'alternate'?

Topic summary

Worksheet Mind map: Angles

- Give 3 examples of where angles are used.
- How confident do you feel about working with angles?
- Is there anything you did not understand? Ask a friend or your teacher for help.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 2

ANSWERS ON P. 567

Quiz
Test yourself 2

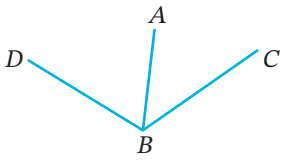
1 Draw and label each angle. 2.01

a $\angle BKT$ **b** $\angle FPR$ **c** $\angle MZQ$

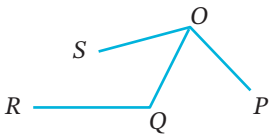
2 Use a protractor to measure each angle you drew in question **1**. Name the smallest angle and the largest angle. 2.02

3 Name the angle adjacent to: 2.01

a $\angle ABC$



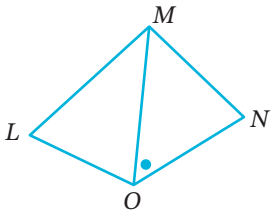
b $\angle POQ$



4 Name the angle marked with a •. 2.01
Select the correct answer **A**, **B**, **C** or **D**.

A $\angle NMO$ **B** $\angle LON$

C $\angle OMN$ **D** $\angle MON$



5 Use a protractor to draw each angle. 2.02

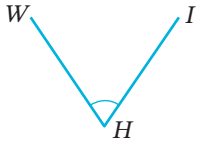
a $\angle JUG = 84°$ **b** $\angle QRA = 117°$ **c** $\angle POT = 41°$

d $\angle DGE = 150°$ **e** $\angle SAR = 96°$ **f** $\angle XDW = 210°$

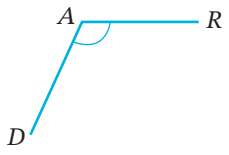
g $\angle MNB = 195°$ **h** $\angle PLO = 270°$ **i** $\angle AMP = 300°$

6 Name each angle using 3 letters, then classify as acute, obtuse, right, reflex or straight. 2.03

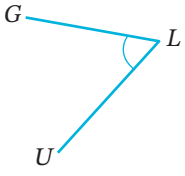
a



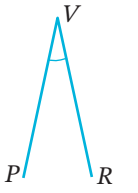
b



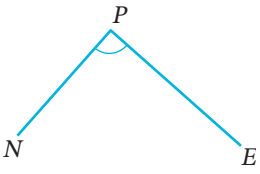
c



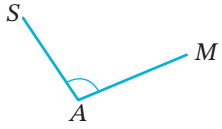
d



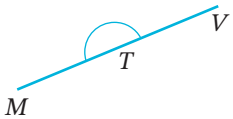
e



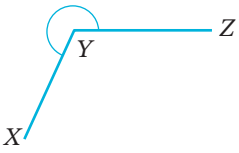
f



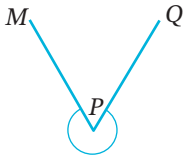
g



h



i



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2.04

7 **a** Write the complement of:

i 35° **ii** 78° **iii** 4°

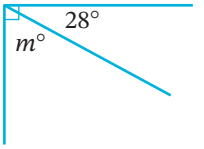
b Write the supplement of:

i 45° **ii** 100° **iii** 178°

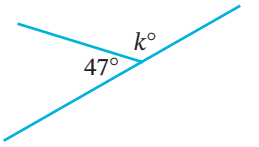
2.04

8 Find the value of each variable.

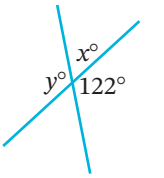
a



b



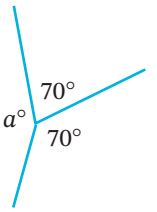
c



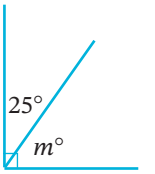
2.05

9 Find the value of each variable.

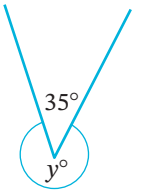
a



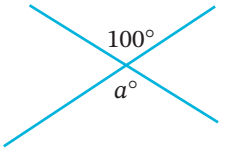
b



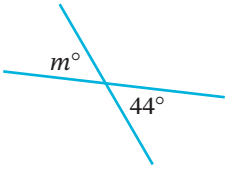
c



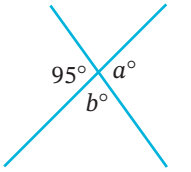
d



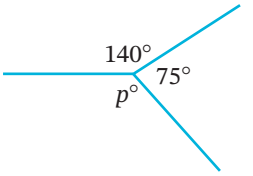
e



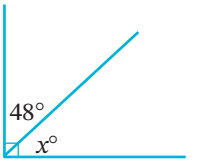
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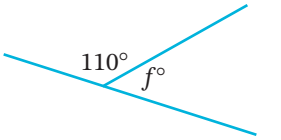
g



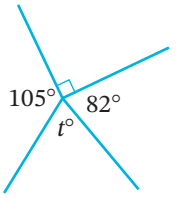
h



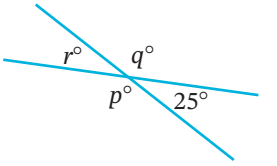
i



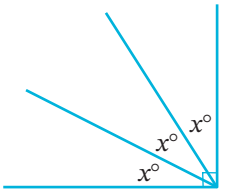
j



k



l



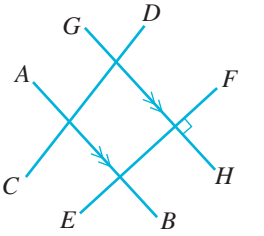
2.06

10 In this diagram, name 2 lines that:

a are parallel

b are perpendicular

c intersect.

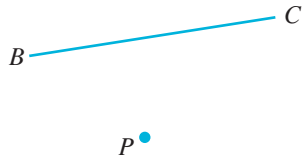


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11 Copy each diagram and use compasses to construct a line parallel to BC going through P. 2.06

a

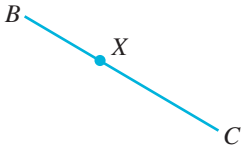


b

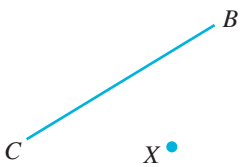


12 Copy each diagram and use compasses to construct a line perpendicular to BC going through X. 2.06

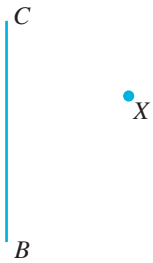
a



b

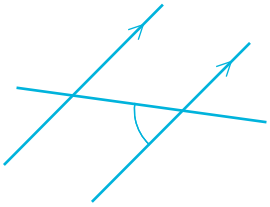


c

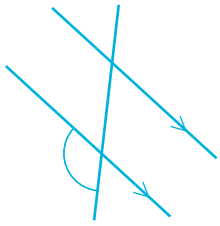


13 Copy each diagram and mark in the corresponding angle to the one shown. 2.07

a

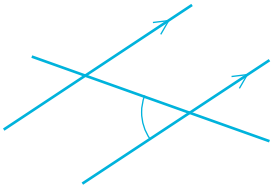


b



14 Copy each diagram and mark in the alternate angle to the one shown. 2.08

a

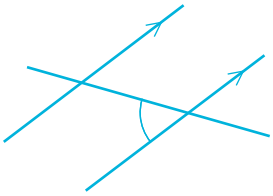


b

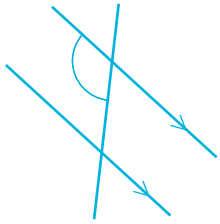


15 Copy each diagram and mark in the co-interior angle to the one shown. 2.09

a



b

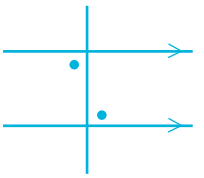


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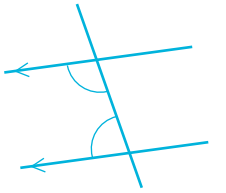
2.10

16 Label the marked pairs of angles as corresponding, alternate or co-interior.

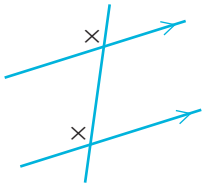
a



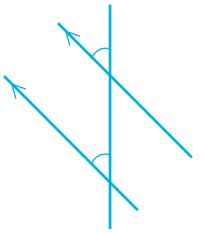
b



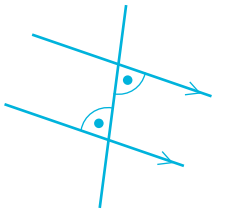
c



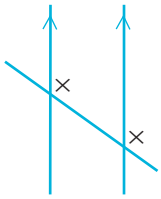
d



e



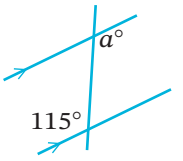
f



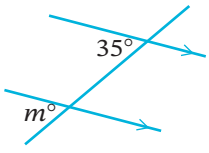
2.10

17 Find the value of each variable.

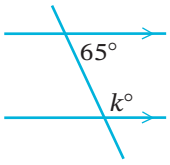
a



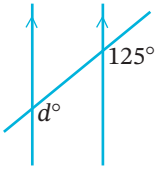
b



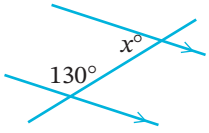
c



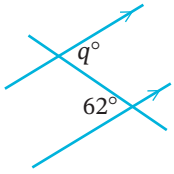
d



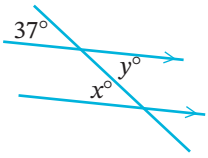
e



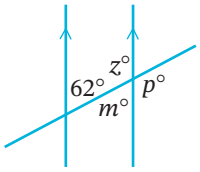
f



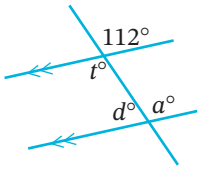
g



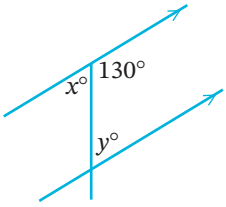
h



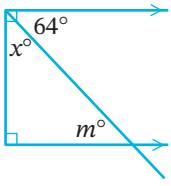
i



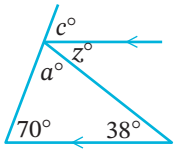
j



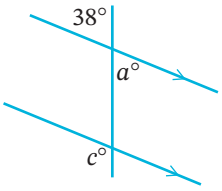
k



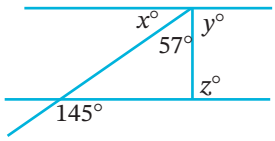
l



m



n

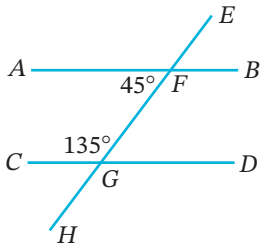


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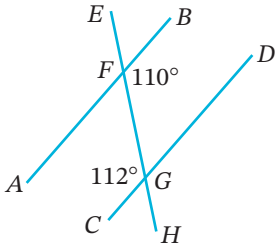
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18 In each diagram below, determine whether $AB \parallel CD$, giving a reason. 2.11

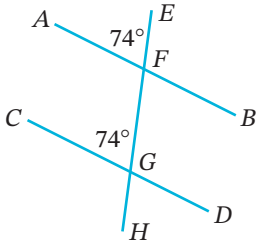
a



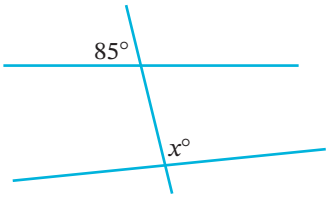
b



c



19 Which value of x would prove that $AB \parallel CD$? 2.11
Select **A**, **B**, **C** or **D**.



A 95 **B** 85

C 275 **D** 5

Foundation Standard Complex